

Macro Asymmetric Trade Costs and Current Account Dynamics

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Abstract

We develop a model-consistent methodology to measure and decompose macro-level export and import costs into common and asymmetric components. These two components yield fundamentally different macroeconomic consequences. Common shocks reduce trade openness but are neutral for external balances. Asymmetric shocks produce significant fluctuations in the trade balance, the current account, and the real exchange rate by distorting the effective price of international lending and pulling absorption forward. We validate these measures using quasi-experimental variation from trade policy events. Tariff increases predominantly generate common cost movements, explaining why protectionism often fails to improve the trade balance.

JEL codes: F10, F13, F14, F32, F41.

Keywords: import costs, export costs, trade cost, tariffs, trade balance, current account, exchange rate , gravity estimation.

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1 Introduction

Whether and how trade restrictions affect the current account and the exchange rate remains a fundamental challenge in international macroeconomics, one that has gained renewed urgency as trade barriers proliferate across major economies (Fajgelbaum et al. 2020; Gopinath et al. 2025). Yet despite its increasing relevance, the profession has not reached a consensus: theoretical models predict that trade cost shocks should affect current account and exchange rate dynamics, while empirical studies find effects that are negligible or statistically indistinguishable from zero.¹ This paper shows that once trade cost shocks are distinguished between common ones, affecting total macro export and import costs equally, and asymmetric ones, affecting them differentially, the disconnect disappears: temporary common shocks affect only trade volumes, while temporary asymmetric shocks have first-order effects on the current account and the real exchange rate.

Two features of trade cost shocks determine their macroeconomic effects: persistence and symmetry. While the role of persistence is now well understood empirically (Schmitt-Grohé et al. 2025), the distinction between common and asymmetric shocks remains unexplored. This distinction is empirically critical. Trade policy changes frequently confound the two: a unilateral tariff that is not retaliated raises import costs asymmetrically, but once retaliation is implemented export costs rise in tandem, generating a common trade cost change. Whether a given policy episode produces common or asymmetric cost movements depends on the extent of retaliation, multilateral liberalization, or reciprocity, making the two in principle indistinguishable from observed policy changes alone. Our key insight is that empirical implementations that look directly at trade policy conflate the two types of shocks, biasing estimated impacts on external balances toward zero. Common shocks affect all countries similarly and provide little incentive for cross-country reallocation of savings or investment.

To overcome this problem, we develop a model-consistent methodology to separately

¹For theoretical and quantitative work, see for example (Rogoff et al. 2000; Itskhoki et al. 2025; Mac Mullen et al. 2023; Alessandria and Choi 2021; Costinot et al. 2025; Jeanne et al. 2024; Fitzgerald 2012; Reyes-Heroles et al. 2016); for empirical work, see (Boz et al. 2019; Furceri et al. 2022; Boer et al. 2024; Barattieri et al. 2021; Schmitt-Grohé et al. 2025).

identify macro-level export and import costs and decompose them into common and asymmetric components. We validate these measures against observed trade policy events, benchmark them against standard indices in the trade literature, and use an instrumental variable approach to establish their macroeconomic relevance. Our empirical results highlight the distinct macroeconomic consequences of common and asymmetric trade cost shocks and the importance of differentiating between them in empirical work as suggested by our theoretical results.

Applying this methodology, we document several facts. Our empirical results shows that temporary trade cost shocks for both common and asymmetric components, yet their macroeconomic consequences differ sharply. Common shocks have negligible effects on the trade balance or the current account. Asymmetric shocks produce significant and persistent external imbalances: a 1% increase in asymmetric trade costs leads to a 0.45 percentage point decline in the trade balance-to-GDP ratio, a 0.48 percentage point reduction in the current account-to-GDP ratio, and a 0.4% real depreciation on impact, equivalent to roughly one quarter of the annual standard deviation of external balances and 8% of that of the REER. The real depreciation persists but is insufficient to fully offset the trade distortion (Jeanne et al. 2024). Real GDP remains largely unaffected, consistent with Schmitt-Grohé et al. (2025) results for temporary tariffs; the deterioration of external balances is instead driven by a rise in both consumption and investment, as asymmetric costs raise the effective price of international lending and pull absorption forward. We also examine the pass-through of tariffs into common versus asymmetric cost components, with implications for the effectiveness of protectionist policies.

Our empirical approach is guided by a multi-country, two-period general equilibrium model with investment and endogenous external balances. The two-period structure is sufficient to derive our main results in closed form (Costinot et al. 2025), while the empirical analysis imposes no restriction on the persistence or dynamics of shocks. We allow for a generalized trade cost structure that features two-sided asymmetric aggregate trade costs, nesting canonical specifications as special cases, including the one-sided asymmetries of Eaton and Kortum (2002) and Waugh (2010) and the symmetric formulations of (Head and Ries 2001; Novy 2013). The model delivers two results. First, common shocks cancel

in the current account equation while asymmetric shocks enter directly. The intuition is that common shocks affect all countries similarly, providing little incentive for cross-country reallocation of savings or investment, whereas asymmetric shocks do. We derive this result in closed form, showing that the net effect of asymmetric shocks depends on the relative strength of three channels: expenditure switching, intertemporal smoothing, and investment reallocation. Since the sign is ambiguous a priori, the question is ultimately empirical. Second, the standard fixed-effects gravity estimator cannot separately identify both directional cost components: it recovers only their product. This last result rules out the use of standard gravity methods for the question at hand, as cannot separately recover aggregate export and import costs a fundamental step for recovering asymmetric trade cost variations in the data.

This result is consistent with the Lerner symmetry predictions: export and import cost changes have isomorphic effects on trade volumes. Nonetheless, these costs have differential effects on relative consumption prices. Combining expenditure shares with relative price indices within a two-country aggregation of each economy against the rest of the world, we separately recover macro-level export and import costs and decompose them into their common and asymmetric components. We show that our identification procedure guarantees the proper aggregation of bilateral trade costs in a model-consistent way, as we establish theoretically; hence, our measures capture in a single statistic both aggregate directional and weighted bilateral trade costs.

We then validate these measures against large tariff changes and Large Asymmetric Macro Trade Policy Events from the IMF’s AREAER database. In both exercises, our measures correctly capture the direction and magnitude of the underlying policy change. Standard gravity-based alternatives fail to detect the direction of the shock.

Having established that our measures track observed policy changes as expected, we turn to estimating their macroeconomic effects. A natural concern is that trade costs, as we find, are partly shaped by policy decisions that may themselves respond to domestic macroeconomic conditions affecting the current account and the exchange rate. Using our measures directly as shocks could therefore bias the estimated relationships. To address this

endogeneity, we construct a Bartik shift-share instrument that exploits countries’ differential exposure to global sectoral cost movements, providing plausibly exogenous variation to any individual country’s domestic conditions.

Literature. This paper makes four contributions. First, we provide the first systematic cross-country empirical evidence that asymmetric trade costs affect current account dynamics through an intertemporal price channel, contributing to the empirical literature on current account determinants (Chinn et al. 2003; Obstfeld et al. 2005; Gourinchas et al. 2007). A growing theoretical literature has shown that asymmetric trade cost movements can generate external imbalances (Barattieri 2022; Rogoff et al. 2000; Erceg et al. 2018; Alessandria and Choi 2021; Mac Mullen et al. 2023; Costinot et al. 2025). We complement these frameworks by providing direct reduced-form panel evidence. Consistent with these frameworks, we find that it is not only the persistence of trade cost shocks that matters for their macroeconomic effects (Schmitt-Grohé et al. 2025) but also their composition into symmetric and asymmetric components, a distinction of fundamental importance when related to trade policy that has been overlooked in the growing body of empirical work studying the macroeconomic effects of trade barriers (Boer et al. 2024; Furceri et al. 2022; Schmitt-Grohé et al. 2025; Estefania-Flores et al. 2025; Barattieri et al. 2021).

Second, we reconcile the tension between quantitative structural models predicting large effects of trade costs on external balances (Alessandria, Bai, et al. 2025; Alessandria and Choi 2021; Rogoff et al. 2000; Erceg et al. 2018; Dekle et al. 2007; Dekle et al. 2008; Eaton, Kortum, et al. 2016; Mac Mullen et al. 2023; Itskhoki et al. 2025) and reduced-form studies consistently finding negligible effects (Boz et al. 2019; Boer et al. 2024; Furceri et al. 2022; Estefania-Flores et al. 2025; Barattieri et al. 2021). The discrepancy stems from a latent measurement error: standard indices conflate symmetric and asymmetric cost movements, attenuating estimated effects toward zero.

Third, we propose a generalized trade cost structure that allows for two-sided asymmetric aggregate trade costs, nesting canonical specifications as special cases, and provide a new methodology to estimate countries’ aggregate export and import costs and to separately decompose them into common and asymmetric components. The existing quantitative trade

literature has relied on restricted formulations: Eaton and Kortum (2002) allow for one-sided importer asymmetries, Waugh (2010) for one-sided exporter asymmetries, and widely used approaches such as Head and Ries (2001) and Novy (2013) impose symmetry altogether. As we formally show, the standard two-way fixed effects gravity estimator cannot separately identify both directional components under our more general structure, recovering only their product. This results is a direct implication from the Lerner symmetry, which implies that export and import cost changes are observationally equivalent in trade volume data, making them impossible to separately identify from trade flows alone. However, these costs have opposite effects on the relative price of the domestic consumption basket (Itskhoki et al. 2025). Combining expenditure shares with relative price indices, we separately recover macro-level export and import costs, advancing the structural estimation of trade costs (Herman 2023; Yotov 2022).

Fourth, we document that tariff increases predominantly generate common rather than asymmetric cost movements. While this need not be the case in principle, our evidence shows that on average, retaliation through tariffs or non-tariff measures causes export costs to rise in tandem with import costs. This offers a new explanation for why tariff changes often fail to improve the trade balance (Erceg et al. 2018; Jeanne et al. 2024), connecting to the growing literature on the macroeconomic effects of trade policy (Fajgelbaum et al. 2020; Schmitt-Grohé et al. 2025).

Layout. The remainder of the paper is organized as follows. Section 2 develops the theoretical framework. Section 3 shows the measurement approach and its mapping to the theory. Section 4 describes the data. Section 5 test and validates the measures. Section 6 presents the main empirical findings. Section 7 examines tariff relationship to common and asymmetric components. Section 8 concludes.

2 Framework

The world economy consists of N countries indexed by $i, j \in \{1, \dots, N\}$, populated by representative consumers with CES preferences over differentiated goods. Time is discrete with two periods, $t \in \{1, 2\}$, as in Costinot et al. (2025). The two-period structure is

sufficient to derive our main results in closed form while capturing the key intertemporal mechanisms through which trade costs affect external balances. In each period, countries trade goods subject to iceberg trade costs, save or borrow through a risk-free international bond, and invest in physical capital. The combination of endogenous savings, investment, and trade cost asymmetries generates endogenous current account imbalances, the central object of our analysis.

Assumption 1. *Preferences.* Country i 's utility is $U^i = u(C_1^i) + \beta u(C_2^i)$, $\beta \in (0, 1)$, where u is twice continuously differentiable, $u' > 0$, $u'' < 0$, and satisfies Inada conditions.

Assumption 2. *Technology.* In period 1, country i receives endowment $y_1^i > 0$ and invests $k^i \geq 0$ units of its own good. In period 2 it produces $f(k^i)$, where f is twice continuously differentiable, $f' > 0$, $f'' < 0$, $f'(0) = \infty$, $f'(\infty) = 0$. Capital depreciates fully.

Assumption 3. *Financial Markets.* A single risk-free bond is traded internationally. The gross return R is determined in equilibrium. We also assume that countries are ex-ante symmetric: $y_1^i = y_1$ and $b_0^i = 0$ for all i .

Optimization and Equilibrium. The representative consumer in country i chooses consumption, savings, and investment to maximize:

$$\max_{C_1^i, C_2^i, b^i, k^i} u(C_1^i) + \beta u(C_2^i) \quad (1)$$

subject to the period budget constraints:

$$P_1^i C_1^i + k^i + \frac{b^i}{R} = p_{i,1} y_1 \quad (2)$$

$$P_2^i C_2^i = p_{i,2} f(k^i) + b^i \quad (3)$$

where P_t^i is the CES price index in country i , $p_{i,i,t}$ is the domestic producer price, b^i denotes net bond holdings (with $b^i > 0$ representing lending), and R is the gross world interest rate.

The current account of country i in period 1 is:

$$CA_1^i = \frac{b^i}{R} = p_{i,1} y_1 - P_1^i C_1^i - k^i \quad (4)$$

which equals the difference between income and total domestic absorption (consumption plus investment). The optimality conditions are:

$$u'(C_1^i) \frac{1}{P_1^i} = \beta R u'(C_2^i) \frac{1}{P_2^i} \quad (5)$$

$$p_{i,2} f'(k^i) = R \quad (6)$$

Equation (5) is the Euler equation equating the marginal rate of substitution to the real interest rate, adjusted for changes in the consumption price index. Equation (6) equates the marginal product of capital to the world interest rate.

Market Clearing. Goods markets clear in each period: total production equals domestic absorption plus net exports. The bond market clears internationally:

$$\sum_{i=1}^N b^i = 0 \quad (7)$$

An equilibrium consists of allocations $\{C_t^i, k^i, b^i\}_{i,t}$, prices $\{p_{i,t}, P_t^i\}_{i,t}$, and an interest rate R such that consumers optimize, goods and bond markets clear, and trade flows satisfy the demand system (8) subject to iceberg costs.

Multi-Country Trade Structure. The representative consumer in country j has CES preferences over domestic and foreign varieties with elasticity of substitution $\sigma > 1$ and home bias parameter φ . Demand for goods from country i in country j is:

$$q_{i,j,t} = \begin{cases} \varphi \left(\frac{p_{j,j,t}}{P_{j,t}} \right)^{-\sigma} C_{j,t} & i = j \\ (1 - \varphi) \left(\frac{p_{i,j,t}}{P_{j,t}} \right)^{-\sigma} C_{j,t} & i \neq j \end{cases} \quad (8)$$

with aggregate price index $P_{j,t} = \left[\varphi p_{j,j,t}^{1-\sigma} + (1-\varphi) \sum_{i \neq j} p_{i,j,t}^{1-\sigma} \right]^{1/(1-\sigma)}$. Trade is subject to iceberg costs: shipping from i to j requires sending $\tau_{i,j,t} \geq 1$ units for one to arrive. We impose the following structure on these costs.

Assumption 4. Generalized Trade Costs. *Total iceberg costs from i to j decompose multiplicatively as:*

$$\tau_{i,j,t} = \tau_{i,t}^X \cdot \tau_{j,t}^M \cdot \tau_{i,j,t}^B \quad (9)$$

where $\tau_{i,t}^X \geq 1$ is an aggregate export cost common to all destinations of i , $\tau_{j,t}^M \geq 1$ is an aggregate import cost common to all origins of j , and $\tau_{i,j,t}^B \geq 1$ is a bilateral component specific to the (i, j) pair. Domestic costs are normalized: $\tau_{i,i,t} = 1$.

Assumption 4 nests the standard specifications as special cases. Eaton and Kortum (2002) and others allows only for aggregate importer cost, implicitly restricting macro export costs to be homogeneous across origins, so that only aggregate destination-specific import barriers vary. Waugh (2010) instead restricts macro import costs to be homogeneous, allowing only aggregate origin-specific export barriers to vary. Symmetric formulations (James E. Anderson et al. 2003; Novy 2013) impose $\tau_{i,j,t} = \tau_{j,i,t}$, collapsing to the Head-Ries index (Head and Ries 2001). Our framework allows both aggregate trade costs, $\tau_{i,t}^X$ and $\tau_{j,t}^M$ to vary freely across countries and over time. Empirically, $\tau_{j,t}^M$ captures frictions uniform across all import origins — such as across-the-board tariffs or foreign exchange surrender requirements — while $\tau_{i,t}^X$ captures frictions common to all export destinations, such as for example export repatriation requirements or export licensing among others. A country can simultaneously tighten import restrictions while relaxing export barriers, precisely the asymmetric policy stance this paper studies.

This two-sided structure has two immediate consequences, formalized below. First, standard structural gravity estimation with two-way fixed effects cannot separately identify $\tau_{i,t}^X$ and $\tau_{j,t}^M$ when both vary freely across countries (Proposition 1). Second, the domestic economy’s aggregate costs are preserved under aggregation to a two-country framework (Proposition 2), enabling us to identify total directional export and import costs by aggregating each country against the rest of the world as we will detail later.

Gravity Estimation Under Generalized Trade Costs. The workhorse approach to estimating trade costs exploits two-way fixed effects on bilateral trade flows, recovering trade cost parameters while absorbing multilateral resistance terms (James E. Anderson et al. 2003; Eaton and Kortum 2002; Head and Mayer 2014; Yotov 2022; Waugh 2010). Starting from the demand system (8) and the pricing condition $p_{i,j,t} = \tau_{i,j,t} p_{i,i,t}$, relative expenditure satisfies:

$$\ln \frac{\exp_{ij}}{\exp_{jj}} = \underbrace{(1 - \sigma) (\ln p_{ii} + \ln \tau_i^X)}_{\gamma_i} + \underbrace{(1 - \sigma) (-\ln p_{jj} + \ln \tau_j^M)}_{\delta_j} + \underbrace{(1 - \sigma) \ln \tau_{ij}^B}_{\epsilon_{ij}} \quad (10)$$

which is estimated as a two-way fixed effects regression $\ln(\exp_{ij}/\exp_{jj}) = \gamma_i + \delta_j + \epsilon_{ij}$. The bilateral residual $\hat{\epsilon}_{ij}$ consistently recovers τ_{ij}^B regardless of the structure of aggregate costs. To construct country-level directional costs, the standard practice combines a country's own fixed effects, $\hat{\gamma}_i + \hat{\delta}_i$, which recovers $\tau_i^X \cdot \tau_i^M$.² Under usual one-sided restrictions $\tau_i^X = 1$ for all i (as in Eaton and Kortum 2002) or $\tau_j^M = 1$ for all j (as in Waugh 2010) this correctly recovers total directional costs, since the unrestricted component is homogeneous. Under Assumption 4, however, both components vary freely: $\hat{\gamma}_i + \hat{\delta}_i$ recovers $\tau_i^X \cdot \tau_i^M$ rather than the required $\tau_i^X \cdot \tau_j^M$, generating a bias of $\ln \tau_i^M - \ln \tau_j^M$. Proposition 1 formalizes this result.

Proposition 1 Bias in Country-Level Trade Cost Estimation. *Under Assumptions 1–4 and CES demand, the two-way fixed effects gravity estimator yields:*

1. *Consistent bilateral trade costs: $\hat{\tau}_{ij}^B \xrightarrow{p} \tau_{ij}^B$ regardless of macro cost heterogeneity.*
2. *Consistent country-level directional costs if and only if there exist constants $\alpha^X, \alpha^M \in \mathbb{R}_+$ such that $\tau_{i,t}^X = \alpha^X$ for all i , or $\tau_{j,t}^M = \alpha^M$ for all j , or both.*

Proof. See Appendix A.1. □

Proposition 1 has two implications. First, it validates the large literature using gravity to study bilateral barriers, such as the effects of distance, language, and trade agreements

²See for example Eaton and Kortum (2002), Waugh (2010), Dekle et al. (2007), and Boz et al. (2019).

(Head and Ries 2001; James E Anderson et al. 2004), as these are robustly identified by $\hat{\epsilon}_{ij}$ even under Assumption 4. Second, it shows that gravity methods cannot separately identify $\tau_{i,t}^X$ and $\tau_{i,t}^M$ when both vary: the fixed effects recover only their product. This is of first-order importance for our analysis, since properly estimating asymmetric and common trade costs requires separately measuring directional export and import costs. This limitation motivates our alternative approach. Before developing the estimator, we first establish that the domestic economy's aggregate trade costs are preserved when aggregating from the multi-country to a two-country framework. This mapping, derived in the next subsection, is the foundation for the identification strategy developed in Section 6.

2.1 Mapping Between Multi-Country and Two-Country Models

Our identification strategy treats each country d as a domestic economy trading with an aggregate rest of the world f . The trade cost structure mirrors Assumption 4 but adapted to the two-country case:

$$\tilde{\tau}_{d,f} = \tilde{\tau}_d^X \cdot \tilde{\tau}_f^M \quad (11)$$

$$\tilde{\tau}_{f,d} = \tilde{\tau}_f^X \cdot \tilde{\tau}_d^M \quad (12)$$

where tildes denote two-country parameters. With a single trading partner, bilateral costs are perfectly collinear with the aggregate components and are therefore absorbed without loss of generality. The key question is whether the multi-country trade costs map consistently into this framework. The following proposition shows that they do: the domestic economy's macro costs are preserved exactly, while the rest of the world's effective costs are trade-weighted aggregations of the underlying multi-country structure.

Proposition 2 Trade Cost Mapping. *The two-country model replicates country d 's total exports and imports from the multi-country model if and only if:*

$$(i) \text{ **Preserved components: } \tilde{\tau}_d^X = \tau_d^X \text{ and } \tilde{\tau}_d^M = \tau_d^M.**$$

(ii) **Export mapping:** The effective ROW import cost satisfies:

$$(\tilde{\tau}_f^M)^{\sigma-1} = \sum_{j \neq d} \frac{s_{d,j} P_j C_j}{EXP_d} \cdot (\tau_{j,t}^M \cdot \tau_{d,j}^B)^{\sigma-1} \quad (13)$$

where $s_{d,j}$ is country j 's expenditure share on goods from d .

(iii) **Import mapping:** The effective ROW export cost satisfies:

$$(\tilde{\tau}_f^X)^{\sigma-1} = \sum_{j \neq d} \frac{s_{j,d}}{s_{ROW,d}} \cdot (\tau_{j,t}^X \cdot \tau_{j,d}^B)^{\sigma-1} \quad (14)$$

where $s_{ROW,d} = \sum_{j \neq d} s_{j,d}$.

Proof. See Appendix A.2. □

Proposition 2 delivers two results. First, the domestic economy's aggregate costs, τ_d^X and τ_d^M , are preserved exactly under aggregation, invariant to how trading partners are grouped. Second, the rest of the world's effective costs are trade-weighted averages of the underlying bilateral and macro frictions, with weights reflecting each partner's importance in d 's export and import baskets, respectively. Together, these results justify treating each country as a domestic economy in a two-country framework: the domestically relevant trade cost components are recovered without distortion, while the rest-of-world costs are aggregated in a model-consistent way. From here on, unless otherwise noted, we work with the two-country model, treating each country as the domestic economy relative to the rest of the world. We refer to the total directional trade costs measured in this framework as macro export and macro import costs.³

2.2 Asymmetric Trade Costs and External Imbalances

Having established how to map multi-country trade costs into a two-country framework, we now show that the differential change between $\tilde{\tau}_{d,f}$ and $\tilde{\tau}_{f,d}$ has direct economic consequences for external imbalances, the real exchange rate, and the terms of trade. Define the

³In terms of the generalized trade cost structure, each macro cost includes both the aggregate directional component and the trade-weighted bilateral costs.

asymmetric component of trade costs as the log ratio of total import to export costs:

$$\Delta\tau_d \equiv \ln \left(\frac{\tilde{\tau}_{f,d}}{\tilde{\tau}_{d,f}} \right) = \ln \tilde{\tau}_d^M + \ln \tilde{\tau}_f^X - \ln \tilde{\tau}_d^X - \ln \tilde{\tau}_f^M \quad (15)$$

A common change in total export and import costs, leaves $\Delta\tau_d$ unchanged, and has no first-order effect on the current account in a symmetric equilibrium, consistent with Rogoff et al. (2000) as we will show in the following proposition.⁴ It is only the asymmetric component $\Delta\tau_d$ that drives external imbalances. The following proposition formalizes this result and characterizes the sign of the current account response.

Proposition 3 Current Account and Asymmetric Trade Costs. *To first order around the symmetric steady state, in the two-country model:*

$$\tilde{b} = \Phi \Delta\tau_d \quad (16)$$

where $\tilde{b} \equiv b/(\bar{P}\bar{C})$ is the normalized current account, $\Delta\tau_d \equiv \hat{\tau}_{fd,1} - \hat{\tau}_{df,1}$ is the asymmetric component of trade costs in log-deviations from the symmetric steady state, and $\Phi \leq 0$ is given by:

$$\Phi = \frac{\lambda \left[(\sigma - 1)(1 - \lambda) + \lambda(1 - \gamma) + \frac{\omega\psi}{\eta} (\gamma(1 - \sigma(1 - \lambda)) - \lambda) \right]}{\psi\xi \left[1 + 4\lambda(1 - \lambda)(\gamma\sigma - 1) - \frac{\gamma\omega\psi}{\eta} \right]} \quad (17)$$

where $\lambda \equiv \frac{(1-\varphi)\bar{\tau}^{1-\sigma}}{\varphi+(1-\varphi)\bar{\tau}^{1-\sigma}}$, $\gamma \equiv -\frac{u''(\bar{C})\bar{C}}{u'(\bar{C})}$, $\eta \equiv -\frac{f''(\bar{k})\bar{k}}{f'(\bar{k})}$, $\omega \equiv \frac{\bar{k}}{y_1}$, $\xi \equiv 1 - \omega$, $\psi \equiv (1 + \bar{R})/\xi$.

Proof. See Appendix A.3. □

Corollary 4 Neutrality of Common Trade Cost Shocks. *A common trade cost shock, defined as $\hat{\tau}_{df,1} = \hat{\tau}_{fd,1}$, implies $\Delta\tau_d = 0$ and hence $\tilde{b} = 0$ to first order around the symmetric steady state. Common shocks affect trade openness but have no first-order effect on the current account.*

⁴This result extends the insight in Rogoff et al. (2000) that symmetric trade cost changes are neutral for the current account to first order. It need not to hold under country ex ante asymmetries; see Reyes-Heroles et al. (2016).

The scalar Φ aggregates three channels. The first is an expenditure switching effect: when import costs rise relative to export costs, demand shifts toward domestic goods, improving the trade balance when $\sigma > 1$, the standard Marshall-Lerner condition (Robinson 1937; Abba P. Lerner 1944). However, its magnitude is governed by $(\sigma - 1)(1 - \lambda)$, which is small when the import share λ is low. The second is an intertemporal smoothing effect: a temporary asymmetric shock makes the consumption basket more expensive today than tomorrow, inducing households to borrow against future income when the income effect dominates ($\gamma > 1$), deteriorating the current account (Obstfeld et al. 1995). The third is an investment reallocation effect: asymmetric costs alter the real exchange rate and terms of trade, temporarily raising the relative return to domestic capital and stimulating current investment, further increasing absorption and widening the deficit (Baxter et al. 1995; Backus et al. 1993). The strength of this channel is governed by $\omega\psi/\eta$, and is larger when the investment share is high or production curvature is low. The last two channels work against the first, and the denominator of Φ captures the additional offsetting response of the exchange rate and terms of trade, which tends to dampen the effect on the current account as it increases.

Notably, all three channels are scaled by λ : as $\lambda \rightarrow 0$ the economy approaches autarky and $\Phi \rightarrow 0$, so that asymmetric trade cost shocks have no effect on external balances. Initial trade openness therefore acts as an amplifying or mitigating factor, consistent with the insight in Rogoff et al. (2000) that trade integration shapes how shocks, in this case asymmetric trade cost shocks, affect the current account. Since the three channels oppose each other, the sign of Φ cannot be determined a priori without restricting preferences, technology, and the level of trade costs. This ambiguity, together with the abstraction inherent in the theoretical framework, motivates our empirical approach. In Section 6, we develop the estimator for the common and asymmetric components and take these predictions to the data, allowing us to further characterize the impact of trade costs over larger time horizons.

3 Measuring Directional Macro Trade Costs

Having established the relevance of estimating asymmetric versus common trade cost shocks, this section provides the theoretical results for identification. We first discuss how we identify directional macro export and import costs, and then establish the identification and existence of their asymmetric and common latent components.

Define the total export cost from country d to f as:

$$\mathcal{X}_{dt} = \left(\frac{1 - \varphi}{\varphi} \right) \tau_{dft} \quad (18)$$

and the corresponding total import cost to country d from f as:

$$\mathcal{M}_{dt} = \left(\frac{\varphi}{1 - \varphi} \right) \tau_{fdt} \quad (19)$$

These statistics aggregate the total directional time-varying iceberg costs τ_{dft} and τ_{fdt} with the structural home-bias parameter φ , capturing the total distortionary burden on exports and imports, respectively.

To estimate these measures, we utilize the intra-temporal expenditure share conditions. Let s_{dft} denote the share of country f 's total expenditure on goods produced in country d , and s_{ddt} the share of country d 's expenditure on its own goods:

$$s_{dft} = \frac{p_{dft} q_{dft}}{P_{ft} C_{ft}} \quad s_{ddt} = \frac{p_{ddt} q_{ddt}}{P_{dt} C_{dt}} \quad (20)$$

Proposition 5 Macro Trade Cost Measurement. *Under CES preferences with elasticity $\sigma > 1$, the macro export and import costs for country d relative to the rest of the world f are given by:*

$$\mathcal{X}_{dt} = \left(\frac{s_{dft}}{s_{ddt}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{ft}}{P_{dt}} \right) \quad (21)$$

$$\mathcal{M}_{dt} = \left(\frac{s_{fdt}}{s_{fft}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{dt}}{P_{ft}} \right) \quad (22)$$

where P_{dt} and P_{ft} denote the final consumption price in countries d and f , respectively.

Proof. See Appendix A.4. □

Proposition 5 shows that macro export and import costs can be recovered from two observable objects: relative expenditure shares and relative consumption price indices. The construction of these objects from bilateral trade, production, and price data is detailed in Section 4.

The primary innovation of our approach, relative to the existing literature (Head and Ries 2001; Eaton and Kortum 2002; Waugh 2010; Boz et al. 2019), is the explicit use of relative final expenditure prices to break the identification problem posed by the Lerner symmetry (Abba P Lerner 1936). Standard approaches focusing exclusively on trade allocations cannot separately identify export and import costs because these frictions yield isomorphic effects on trade volumes, as established in Proposition 1. However, even in environments where Lerner symmetry holds, the relative price of total expenditure remains sensitive to the directionality of trade distortions. We leverage this property, as noted in Itskhoki et al. (2025). Given our aggregation results established in Proposition 2, our measures extend to the aggregate level the bilateral or sectoral bilateral decompositions used in previous work (e.g. Reyes-Heroles et al. 2016; Fitzgerald 2012; Eaton, Kortum, et al. 2016), achieving separate identification of macro export and import costs.

Identification Example. To illustrate, consider country d imposing a unilateral tariff fully passed through to border prices. This triggers two effects: a direct effect reducing the import share $s_{f dt}$, and a general equilibrium effect raising P_{dt} relative to P_{ft} , which simultaneously affects both $s_{f dt}$ and $s_{d ft}$. An estimator based solely on trade shares would therefore erroneously attribute the tariff to changes in export costs. Relative prices are key to identification: the term $(s_{d ft}/s_{d dt})^{1/(1-\sigma)}$ co-moves one-for-one with (P_{ft}/P_{dt}) under standard assumptions, canceling the endogenous price response and isolating the direct cost change.

Interpretation and Limitations. Our measures are sufficient statistics capturing the wedge required to rationalize observed divergences between international and domestic flows, under the assumption of equal preferences across countries and without requiring as-

assumptions on the production environment, as in standard gravity-based approaches (Eaton and Kortum 2002; Waugh 2010). They encompass any friction that differentially affects cross-border relative to domestic trade, including customer capital accumulation, exporter entry costs, and financial frictions (Fitzgerald et al. 2024; Alessandria and Choi 2021; Kohn et al. 2016), rather than isolating trade policy alone. This broad scope ensures complete accounting of the total distortionary burden, at the cost of not decomposing the wedge into specific sub-components. The decomposition into common and asymmetric components developed below allows us to isolate the macroeconomically relevant variation.

Average Trade Costs and the Head-Ries Index. Our measures provide a natural generalization of the Head-Ries (HR) index (Head and Ries 2001). By taking the geometric average of our export and import cost statistics, we recover the symmetric HR index:

$$\tau_t^{HR} = (\mathcal{M}_{dt}\mathcal{X}_{dt})^{\frac{1}{2}} = \left(\frac{s_{f dt}s_{d ft}}{s_{f ft}s_{d dt}} \right)^{\frac{1}{2(1-\sigma)}} \quad (23)$$

While structurally consistent with our framework under symmetry, the HR index lacks the dimensionality to separate directional shocks to export and import costs. By collapsing these costs into a geometric mean, it captures only the common component of trade frictions.

3.1 Decomposition: Common and Asymmetric Components

We now decompose these measures into their common and asymmetric latent components, whose distinct macroeconomic effects were established in Proposition 3. We define the common component, $\tau_{d,t}^C$, as the source of symmetric fluctuations in trade costs, and the asymmetric component, $\tau_{d,t}^A$, as the source of directional divergence. To formalize this decomposition, let $\varepsilon_{z,\tau^j} \equiv \frac{\partial \ln z_{d,t}}{\partial \ln \tau_{d,t}^j}$ denote the elasticity of trade cost measure $z \in \{\mathcal{X}, \mathcal{M}\}$ with respect to latent shock τ^j for $j \in \{A, C\}$.

Definition 1 (Common Trade Costs). A shock $\tau_{d,t}^C \in \mathbb{R}_+$ is defined as common if it satisfies:

$$\varepsilon_{\mathcal{X}_{d,t},\tau_{d,t}^C} = \varepsilon_{\mathcal{M}_{d,t},\tau_{d,t}^C} = \varepsilon^C \quad \forall \{d,t\}$$

where $\varepsilon^C \in \mathbb{R}_+$ is a constant scaling parameter.

Definition 2 (Asymmetric Trade Costs). A shock $\tau_{d,t}^A \in \mathbb{R}_+$ is defined as asymmetric if it satisfies:

$$\varepsilon_{\mathcal{X}_{d,t}, \tau_{d,t}^A} - \varepsilon_{\mathcal{M}_{d,t}, \tau_{d,t}^A} = \varepsilon^A \quad \forall \{d, t\}$$

where $\varepsilon^A \in \mathbb{R}_+$ represents the magnitude of the differential impact on export versus import costs; the sign of the directional effect is absorbed into $\tau_{d,t}^A$.

Proposition 6 Decomposition of Trade Costs. Let $\mathcal{X}_{d,t}$ and $\mathcal{M}_{d,t}$ be positive observable trade cost measures. There exist latent components $\tau_{d,t}^C, \tau_{d,t}^A \in \mathbb{R}_+$ and mapping functions $f, g : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ such that:

$$\mathcal{X}_{d,t} = f(\tau_{d,t}^C, \tau_{d,t}^A) \quad \text{and} \quad \mathcal{M}_{d,t} = g(\tau_{d,t}^C, \tau_{d,t}^A)$$

subject to the elasticity constraints in Definitions 1 and 2, if and only if:

- (i) **Smoothness:** $f, g \in \mathcal{C}^2(\mathbb{R}_+^2)$ are twice continuously differentiable.
- (ii) **Non-collinearity:** $\mathcal{X}_{d,t}$ and $\mathcal{M}_{d,t}$ are not perfectly collinear across the sample.

Proof. See Appendix A.5. □

Corollary 7 Scale Identification. The mapping between latent components and observable trade costs is identified up to a scaling constant.

Proof. See Appendix A.5. □

Two implications follow from these results. First, since our analysis focuses on within-country time variation, identification up to scale does not pose an empirical challenge. Second, the HR index, recovered as the special case where $\varepsilon_{\mathcal{X}, \tau^C} = \varepsilon_{\mathcal{M}, \tau^C} = 1$, is by construction orthogonal to the asymmetric component and therefore an insufficient statistic for analyzing the macroeconomic consequences of trade costs on external imbalances, as established in Proposition 3.

4 Data and Empirical Measurement

This section describes the construction of our export and import cost statistics and details the data sources utilized in our empirical analysis. Our primary measurement challenge lies in the general unavailability of cross-sectional price levels ($P_t^{j,C}$) that are consistent across countries and over time.

4.1 Construction of Trade Cost Indices

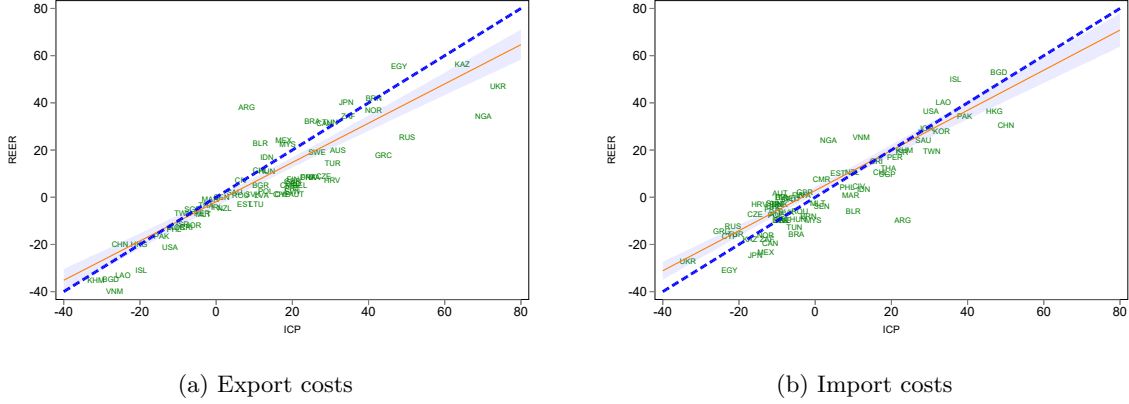
To circumvent the data limitations regarding price levels, we focus on the temporal evolution of trade costs within countries. We construct a time-series index for country d relative to the rest of the world f , normalizing all measures to a base year b , which in practice we set to the initial year in our sample (1996) with full coverage of countries. Our empirical counterparts for macro export and import costs are defined as follows:

$$\mathcal{X}_t^d = \left[\frac{\left(\frac{s_{dft}}{s_{ddt}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{ft}}{P_{dt}} \right)}{\left(\frac{s_b^{d,f}}{s_{ddb}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{fb}}{P_{db}} \right)} \right] \times 100 ; \quad \mathcal{M}_t^d = \left[\frac{\left(\frac{s_{fdt}}{s_{fft}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{dt}}{P_{ft}} \right)}{\left(\frac{s_{fdb}}{s_{ffb}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{db}}{P_{fb}} \right)} \right] \times 100$$

All the terms in previous equation are observable. We utilize REER data to proxy for relative price movements between country d and its trading partners. The trade elasticity is assumed to be equal to 4 for computation (Simonovska et al. 2014).

Despite the efforts to avoid the REER computation to diverge from the ideal consumption price index (Bayoumi et al. 2006) there might be inconsistencies biasing our estimates. We check for this by comparing REER-based measures against estimates from the International Comparison Program (ICP) database (World Bank 2011; Waugh 2010), which a rigorous benchmark for cross-country price comparability with publications scatter across several years. Despite its limited frequency and coverage precluding its use as our primary series, Figure 1 shows strong co-movement between the two sets of estimates, supporting the REER as a robust proxy for the time-varying components of trade distortions.

Figure 1. Trade Cost Differences: REER versus ICP



Note: This figure illustrates the change in estimated export and import costs between 2011 and 2017 using two alternative price proxies. The vertical axis depicts trade costs derived from the Real Effective Exchange Rate (REER), while the horizontal axis utilizes prices from the International Comparison Program (ICP) database.

Computation of Common and Asymmetric Trade Costs. Proposition 6 established that several empirical specifications can identify the common and asymmetric components of trade costs. We prioritize a decomposition that ensures structural interpretability by focusing on a parameterization where $\varepsilon^C = 1$ and $\varepsilon^A = 1$, which allows for a direct mapping between the estimated cost components and the log-linearized trade equations.

To operationalize we follow the case described in proposition 5 proof in appendix A. We estimate the common component as a country-specific geometric mean between the normalized export and import costs. Conversely, we identify the asymmetric component by calculating the log ratio of these aggregate costs, normalized to a 1996 baseline:

$$\tau_{d,t}^A = \left[\ln \left(\frac{\mathcal{X}_{d,t}}{\mathcal{M}_{d,t}} \right) - \ln \left(\frac{\mathcal{X}_{d,1996}}{\mathcal{M}_{d,1996}} \right) \right] \times 100 \quad (24)$$

Using the normalized measures ensures that both common and asymmetric trade costs represent the percentage-point change relative to the start of the sample.

4.2 Data and Variable Construction

Expenditure Shares. We utilize the OECD Inter-Country Input-Output (ICIO) Tables (1995–2020) to construct the directional expenditure shares. Following Waugh (2010), we

define total nominal expenditure for country d as gross production adjusted for the trade balance. Let Y_{dt} , EX_{dt} , and IM_{dt} denote nominal gross production, total exports, and total imports, respectively. The share of expenditure in country d on goods from the rest of the world is:

$$s_{dft} = \frac{IM_t^d}{Y_t^d - EX_t^d + IM_t^d} \quad (25)$$

The corresponding domestic expenditure share is:

$$s_{ddt} = \frac{Y_t^d - EX_t^d}{Y_t^d - EX_t^d + IM_t^d} \quad (26)$$

For constructing our IV approach, which will be detailed later, we also rely on industry level data, which is sources and treated in the same way as the aggregate one.

Price Indices and Macroeconomic Aggregates. To proxy relative price movements, we use the REER measures from Darvas (2021), which extend the methodology of Bayoumi et al. (2006).⁵ For aggregate variables we use data from the IMF World Economic Outlook (WEO) database, providing consistent measures for GDP, consumption, the trade balance, and the terms of trade. Our final balanced panel includes 72 countries, with the sample size primarily determined by the intersection of ICIO and WEO coverage.

Trade Policies and Tariffs. To identify policy-driven shocks, we rely on two primary sources. We use the IMF’s *Annual Report on Exchange Arrangements and Exchange Restrictions* (AREAER) to construct an index of asymmetric trade restrictions, capturing non-tariff barriers and regulatory shifts that tariff data typically omit. A detailed discussion of the index construction is provided in Section 5. To proxy for aggregate tariffs, we obtain data on import-related tax revenues from the OECD Global Revenue Statistics Database, computing the *effective tariff rate* as the ratio of total import tax revenue to total imports. The data is constructed following Schmitt-Grohé et al. (2025) and Irwin (2000).

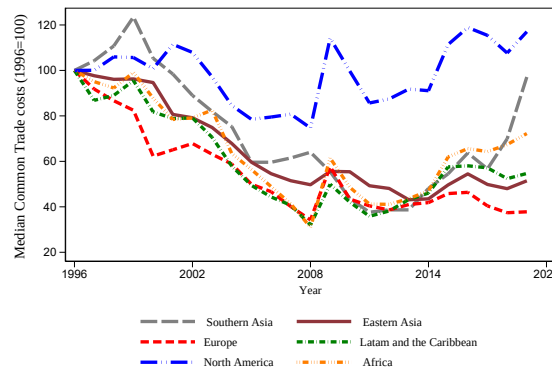
⁵ Bayoumi et al. (2006) construct trade shares that explicitly account for the degree of product differentiation; for differentiated goods, their framework incorporates both direct and indirect competition to ensure the weights reflect the full spectrum of multilateral trade linkages.

4.3 The Evolution of Macro Trade Costs

Before proceeding to the formal validation of our measures, we present the temporal evolution of the common and asymmetric components of trade costs. This provides a baseline for understanding the global trade environment and checks our measures against well-documented historical events.

The Global Dynamics of Common Trade Costs. Figure 2 displays the regional medians of the common component of trade costs. Two distinct patterns emerge. First, the period between 1996 and 2008 was marked by a decline in common trade costs globally. For regions such as North America, Latin America, and Africa, this downward trend accelerated significantly following China’s accession to the WTO in 2001.

Figure 2. Evolution of Common Trade Costs by Region

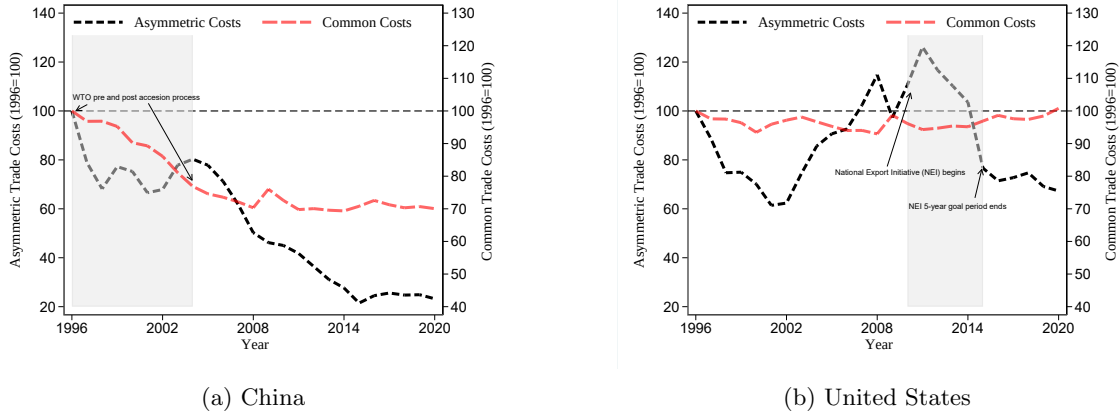


Note: The figure shows the median common trade cost measure for different regions. The trade elasticity is set to $\sigma = 4$.

Second, the secular decline in trade costs stalled abruptly following the Global Financial Crisis (GFC). This spike and subsequent stagnation are consistent with the literature on the Great Trade Collapse and the ensuing slowdown (Bems et al. 2013). Since 2015, we observe a reversal, with common trade costs increasing across most regions. This uptick likely reflects the rising tide of protectionism and the restructuring of global value chains that characterized the late 2010s, trends that appear to have accelerated with increasing geopolitical tensions (Fajgelbaum et al. 2020; Gopinath et al. 2025).

Asymmetric Trade Costs and Policy Divergence. Figure 3 contrasts the evolution of the common and asymmetric trade cost components for China and the United States. In the Chinese case, common trade costs exhibit a sharp decline around WTO accession, followed by a spike during the 2009 Global Financial Crisis. The asymmetric component, however, reveals a distinct pattern. Following a period of relative stability, it experienced a marked decrease beginning in 2006, aligning with the strategic pivot toward domestic import substitution documented by Alessandria, Choi, and Lu (2017).

Figure 3. Asymmetric Trade Costs



Note: The figure shows the asymmetric trade cost measure (dashed line) and the common trade cost (dot-dash line, right-hand axis) for China and the United States. The trade elasticity is set to $\sigma = 4$.

The US experience provides a striking counterpoint. Between 2001 and 2010, while common trade costs were declining, asymmetric trade costs trended upward, indicating that export frictions were rising relative to import costs. This trend reversed in 2010, coinciding with the implementation of the National Export Initiative (NEI), which aimed to double US exports by 2014 through the removal of trade barriers and increased export financing.⁶ These episodes illustrate that the common and asymmetric components capture distinct sources of variation, with the former tracking global trade integration and the latter reflecting country-specific policy orientations.

⁶ See the executive order [here](#).

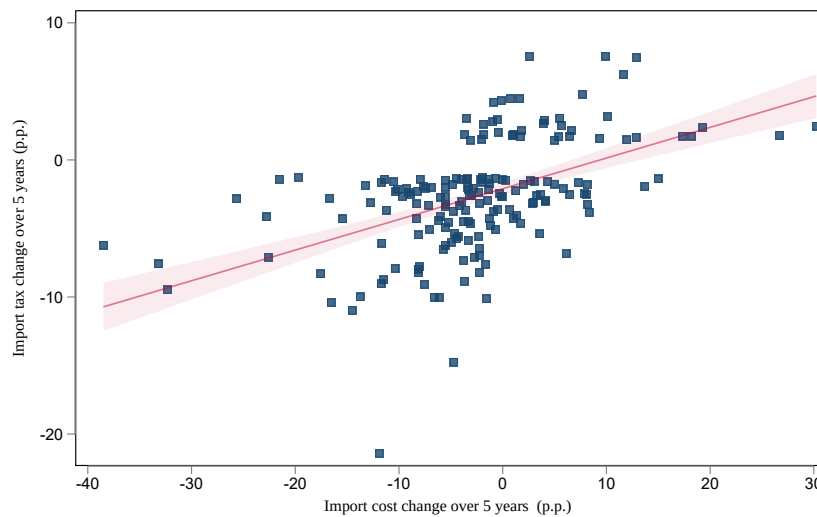
5 Trade Policy and Trade Cost Validation

We now validate our structural trade cost measures against observed trade policy. Since our identification relies exclusively on within-country temporal variation, policy shifts provide a direct test of whether our measures capture the intended distortions. We examine two complementary exercises: the sensitivity of import costs to tariff adjustments, and the response of asymmetric costs to directional policy events.

5.1 Validation 1: Tariffs and Import Costs

Figure 4 plots five-year changes in effective tariffs against the corresponding changes in our estimated import costs across the full sample. Two salient patterns emerge. First, we find a robust and statistically significant positive correlation between tariff movements and our recovered import cost measure, confirming that it captures the intended policy signal. Second, the estimated elasticity exceeds unity, suggesting that tariff movements are accompanied by correlated shifts in non-tariff barriers.

Figure 4. Changes in Import Costs versus Tariffs

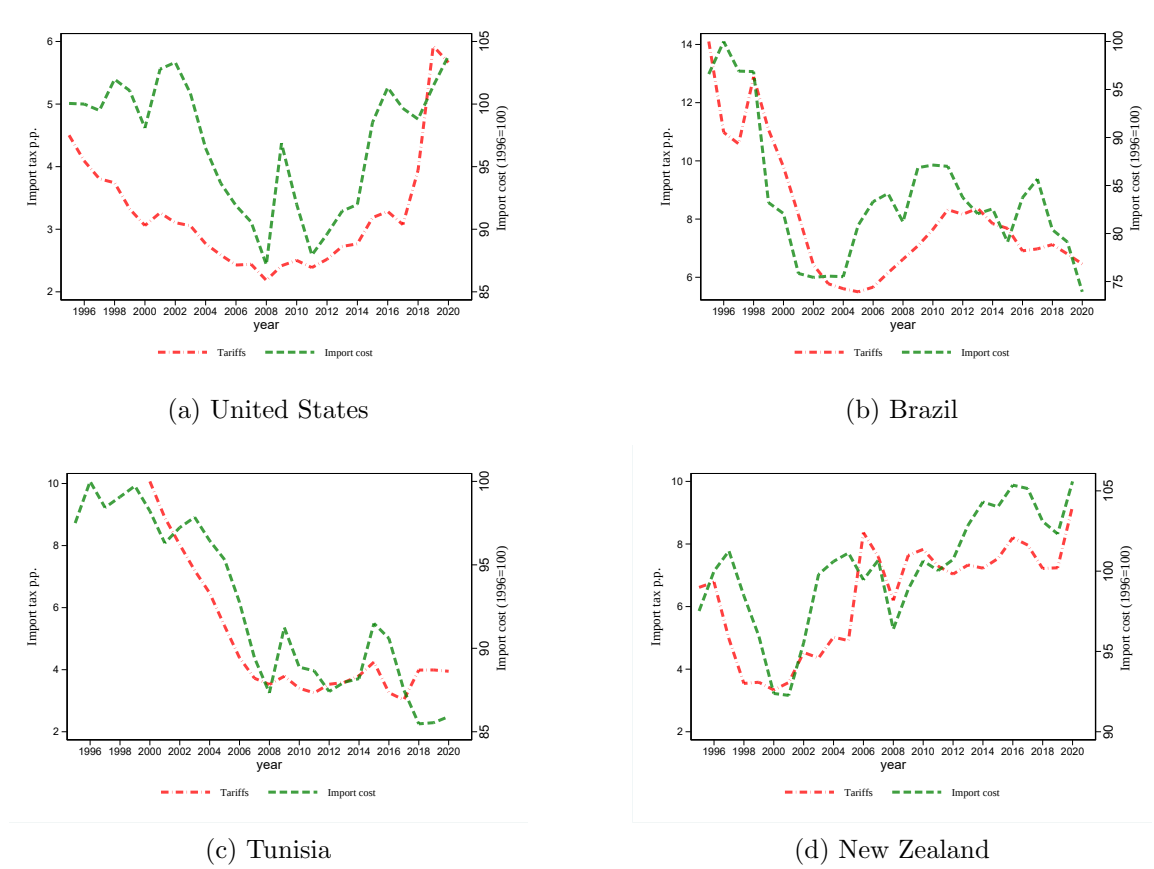


Note: Five-year cumulative changes. Confidence intervals at the 90% level.

Our import cost measure also captures event-window and year-on-year dynamics with

high precision during periods of substantial tariff adjustments. Figure 5 illustrates this alignment for a selection of significant trade liberalization and protectionist episodes. Additional episodes of large tariff changes, defined as those with a cumulative absolute change in aggregate tariffs exceeding 2 percentage points over five years, are detailed in Figure B.2 in the Appendix. Across these diverse institutional contexts, import costs track effective tariffs closely, reinforcing our confidence in using the measure to recover the latent common and asymmetric cost components.

Figure 5. Effective Tariffs versus Import Costs



Note: Evolution of effective tariffs (red dot-dashed, left axis) and import costs (green dashed, right axis) for countries with cumulative tariff changes exceeding two percentage points over five years. Effective tariffs are computed as import tax revenues over total merchandise imports.

5.2 Validation 2: Large Asymmetric Trade Policy Events and Asymmetric Trade Costs

We now evaluate whether our asymmetric trade cost measure responds to directional non-tariff policy shifts, using Large Asymmetric Trade Policy Events (LATPE) as a natural test of structural validity.

We draw on the IMF’s AREAER database (1999–2022), which provides binary indicators of regulatory changes governing international payments and receipts, to isolate episodes of significant directional policy tightening or liberalization.

Quantifying Asymmetric Policy Shocks. We focus on the regulatory categories governing *Exports and Export Proceeds* (Section VIII) and *Imports and Import Payments* (Section VII). For exports, we aggregate five binary variables capturing distortions due to: (1) repatriation requirements; (2) financing requirements; (3) documentation requirements; (4) export licenses; and (5) export taxes. For imports, we aggregate six categories: (1) foreign exchange budgeting; (2) financing requirements; (3) documentation requirements; (4) import licenses and other non-tariff measures; (5) import taxes and/or tariffs; and (6) state import monopolies.⁷

For each country-year pair, these categories define the total number of policies distorting exports ($N_{i,t}^{exp}$) and imports ($N_{i,t}^{imp}$). We construct the *Policy Asymmetry Index*:

$$\text{Policy Asymmetry}_{i,t} = \frac{1 + N_{i,t}^{exp}}{1 + N_{i,t}^{imp}} \quad (27)$$

A LATPE is identified as an episode where $\Delta \text{Policy Asymmetry}_{i,t}$ exceeds the 25th percentile of the distribution of all positive index changes. We formalize these episodes via a treatment dummy $I_{i,t}^{LATPE}$, which equals one at the inception of the shock (T) and remains active for $h > 0$ as long as $\Delta \text{Policy Asymmetry}_{i,T+h} \geq 0$. Table B.1 lists the events included in the estimation. The control group consists of countries exhibiting no changes in the Policy Asymmetry Index throughout the sample ($\Delta \text{Policy Asymmetry}_{i,t} = 0 \forall t$).

⁷This aggregation follows the taxonomic approach of Estefania-Flores et al. (2024) but is restricted to trade-distorting measures.

Estimation. We estimate the dynamic impact using the LP-DiD framework of Dube et al. (2023), which accommodates staggered treatment timing and avoids negative-weighting biases in heterogeneous panels:

$$Y_{i,t+h} - Y_{i,t-1} = \beta^h \Delta I_{i,t}^{LATPE} + \sum_{k=1}^4 \Phi_{h,k} \mathbf{X}_{i,t-k} + \gamma_t + \epsilon_{i,t+h} \quad (28)$$

where $\Delta I_{i,t}^{LATPE}$ is the treatment indicator, $\mathbf{X}_{i,t-k}$ includes four lags of the dependent variable, the log of domestic-to-foreign expenditure shares, the REER, and the terms of trade, and γ_t denotes year fixed effects. The coefficients $\{\beta^h\}$ trace the impulse response relative to the pre-treatment period.

We focus on four outcome variables. First, the log trade ratio ($\ln(EX_{i,t}/IM_{i,t})$), to verify whether LATPEs induce asymmetric shifts in trade flows. Second, we compare three measures of asymmetric trade costs against the policy events:

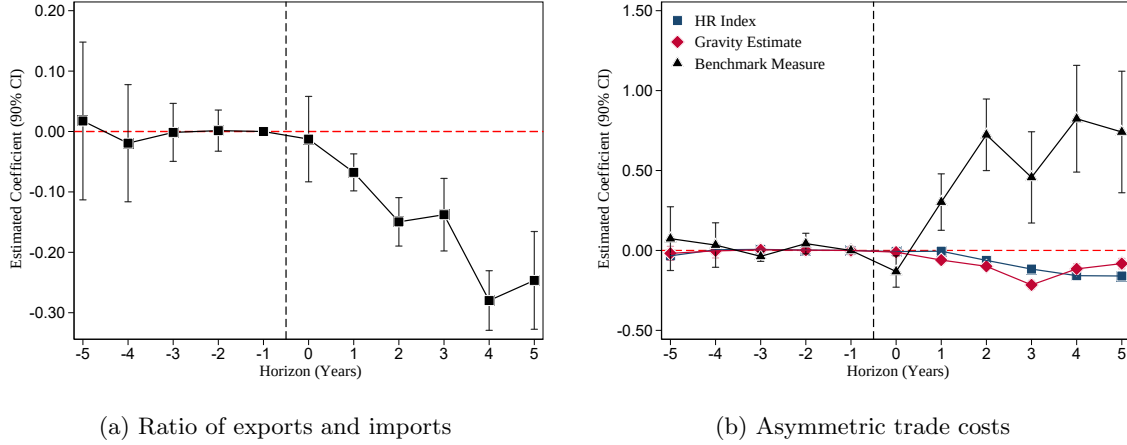
1. **Aggregated Head-Ries (HR) Index:** bilateral trade costs aggregated using the HR index.
2. **Gravity-based estimates:** directional export and import costs from a standard two-way fixed effects gravity specification.
3. **Our aggregate measure:** derived from the structural decomposition of aggregate prices and expenditure shares.

Detailed derivations and aggregation procedures for the first two measures are provided in Appendix B.1.

Results. Figure 6 presents the impulse responses following a LATPE. The absence of pre-trends indicates that policy changes were unrelated to prior trade or trade balance dynamics, supporting the validity of our identification.

Panel (a) illustrates the substantial real-world impact of these events: a LATPE induces a persistent and statistically significant decline in the trade ratio of approximately 30% over a five-year horizon, confirming that our identification strategy captures policy changes with

Figure 6. Event Study Results



Note: LP-DiD estimates from equation (28). LATPE defined as in Section 5.2. Controls: log domestic-to-foreign expenditure, REER, terms of trade, and four-year lags. 90% confidence intervals; standard errors clustered by country.

major macroeconomic consequences in the expected direction.

Panel (b) provides a key result of this paper. Two-way fixed effects gravity estimates (red diamonds) and the aggregated HR index (blue squares) fail to track the policy shock, paradoxically showing a reduction in asymmetric trade costs. In contrast, our aggregate measure (black diamonds) exhibits a robust increase of nearly 70 log points, internally consistent with the observed collapse in the trade ratio and directly capturing the cumulative impact of regulatory frictions that bilateral aggregations miss. These results confirm the identification failure documented in Proposition 1: standard gravity-based aggregations, by recovering only the product of directional costs, cannot detect asymmetric policy shifts.

6 Macroeconomic Consequences of Trade Costs

We now quantify how common and asymmetric trade cost shocks affect external balances, exchange rates, and output.

6.1 Empirical Strategy: Local Projections

We estimate the following local projection using long differences for each horizon h :

$$\Delta_h \ln Y_{i,t+h} = \beta^h \ln(\tau_{i,t}^J) + \Gamma_h' \mathbf{X}_{i,t} + \gamma_t + \gamma_i^{GFC} + \epsilon_{i,t+h} \quad (29)$$

where $\Delta_h \ln Y_{i,t+h} = \ln Y_{i,t+h} - \ln Y_{i,t-1}$ is the cumulative change in the dependent variable from $t-1$ to $t+h$. The coefficients $\{\beta^h\}_{h=1}^{10}$ capture the cumulative elasticity of Y with respect to trade cost measure $\tau_{i,t}^J$ for $J \in \{A, C\}$. Individual fixed effects are absorbed through the long differencing. Proceeding this way also has econometric advantages as it significantly attenuates potential bias from small samples (Piger et al. 2025; Jordà et al. 2025).

The control vector $\mathbf{X}_{i,t}$ includes the terms of trade, the REER, relative domestic-to-foreign expenditure shares, and lagged values of the trade cost measure, with up to four lags throughout. Year fixed effects γ_t absorb global shocks. A country-specific GFC dummy, constructed as the year 2009 interacted with country fixed effects, controls for the idiosyncratic impact of the Great Trade Collapse on individual economies.

Identification: Bartik Instruments. Estimating equation (29) directly may entail significant bias. For instance, governments may change trade restrictions in response to declining productivity or adverse foreign demand shifts, which also affect the current account and exchange rates. We address this potential endogeneity by constructing a shift-share instrument $z_{i,t}$. We apply this instrument specifically to the asymmetric trade cost component, as it is our primary object of interest for identifying intertemporal reallocations.

The instrument is constructed using a leave-one-out strategy at the sectoral level. First, we calculate the average export and import costs for each sector-country pair (s, i) , denoted $\mathcal{X}_{s,i,t}$ and $\mathcal{M}_{s,i,t}$ following previous strategy.⁸ Each captures the trade costs that sector s in country i faces when exporting to or importing from the rest of the world. Given the sectoral-country pair, we construct the average export and import costs from the rest of the

⁸ Sectoral trade costs follow the aggregate approach using sectoral expenditure shares, omitting the price ratio due to data unavailability at the sectoral level.

world excluding country i (denoted $\widehat{\mathcal{X}}_{s,i,t}$ and $\widehat{\mathcal{M}}_{s,i,t}$) as follows:

$$\widehat{\mathcal{X}}_{s,i,t} = \frac{1}{N-1} \sum_{n \neq i} \mathcal{X}_{s,n,t} \quad \text{and} \quad \widehat{\mathcal{M}}_{s,i,t} = \frac{1}{N-1} \sum_{n \neq i} \mathcal{M}_{s,n,t} \quad (30)$$

Second, we aggregate these sectoral costs using country-specific weights based on each sector's share of total trade five years prior.⁹ To satisfy the exclusion restriction, we map the average import costs of the rest of the world to country i 's export costs, reflecting the structural identity that one country's export friction is its partners' import distortion. The resulting instrument is:

$$z_{i,t} = \frac{\sum_s \pi_{s,i,t-5}^{exp} \cdot \widehat{\mathcal{M}}_{s,i,t}}{\sum_s \pi_{s,i,t-5}^{imp} \cdot \widehat{\mathcal{X}}_{s,i,t}} \quad (31)$$

where $\pi_{s,i,t}^{exp}$ and $\pi_{s,i,t}^{imp}$ denote country i 's export and import shares for sector s in year t . The cross-mapping ensures that the instrument reflects cost pressures faced by country i 's trading partners rather than domestic policy choices, satisfying the exclusion restriction under the assumption that global sectoral cost trends are exogenous to any individual country's macroeconomic conditions.

First-Stage Relevance. We estimate the following first-stage specification:

$$\ln(\tau_{i,t}^J) = \alpha + \beta^{IV} \ln(z_{i,t}) + \Gamma' \mathbf{X}_{i,t} + \gamma_t + \gamma_i^{GFC} + \nu_{i,t} \quad (32)$$

where controls are identical to those in equation (29). Table B.3 reports Kleibergen-Paap F-statistics from different time windows. The statistic ranges from 59 to 75 across horizons, confirming strong first-stage relevance.

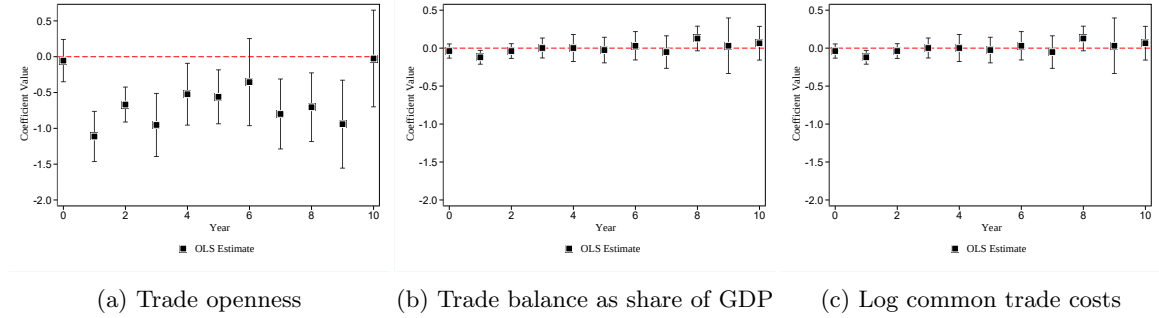
6.2 Main Empirical Findings

We now present our main empirical results. We first analyze the impact of common trade costs before examining the role of asymmetric trade costs in shaping openness and the trade balance. Our analysis reveals five key findings.

⁹ We use five-year lags as our OLS estimates indicate that the macroeconomic effects of asymmetric trade shocks typically dissipate within a five-year horizon.

Fact 1: Common trade costs suppress trade openness but are neutral for the trade balance. Panel (a) of Figure 7 shows that a 1% increase in common trade costs reduces trade openness (defined as exports plus imports over GDP) by around 1 percentage point, as expected. Panel (b) shows that this contraction leaves the trade balance unaffected: common costs impede the level of trade without altering the cross-country saving and investment incentives that determine the current account, consistent with our theoretical results.¹⁰ Panel (c) confirms that common trade cost shocks are temporary despite generating persistent effects on openness. Together, these results highlight the empirical importance of distinguishing between common and asymmetric trade cost shocks.

Figure 7. Common Trade Cost Shocks



Note: OLS estimates. Controls: log terms of trade, log REER, log relative expenditures, four-year lags of all controls and dependent variable. 90% confidence intervals; standard errors clustered by country.

Fact 2: Asymmetric trade costs exhibit significant persistence and mean reversion. Panel (a) of Figure 8 documents the dynamics of the asymmetric cost ratio following a unit shock, normalized to one on impact. The ratio exhibits hump-shaped dynamics, continuing to rise for two years before reverting toward baseline by year five. This temporary increase in the cost of exporting relative to importing generates important intertemporal reallocations. To gauge magnitudes, the annual change in asymmetric costs averages -0.33% with a mean absolute value of 9.6% , and exhibits standard deviations of 13.1% and 8.0% for the annual change and its absolute value, respectively.¹¹

¹⁰Our estimates reflect the cross-country average and do not capture the tilting effect of trade costs through global interest rate fluctuations discussed in Reyes-Heroles et al. (2016).

¹¹Based on changes winsorized at the 1% level, using the same sample as the first horizon of the local projection.

Fact 3: Asymmetric trade costs induce a significant, though temporary, real depreciation. Panel (b) shows that a 1% increase in the asymmetric cost ratio triggers an immediate real depreciation of 0.4%, roughly 8% of the standard deviation of annual REER changes, persisting for three to five years, consistent with the shock dynamics documented in Fact 2. The depreciation arises because asymmetric costs raise export prices relative to import prices, reducing the relative price of the domestic consumption basket (Mac Mullen et al. 2023; Alessandria and Choi 2021). While the REER adjusts in the expected direction, the magnitude is insufficient to insulate the trade balance, as Fact 4 documents.

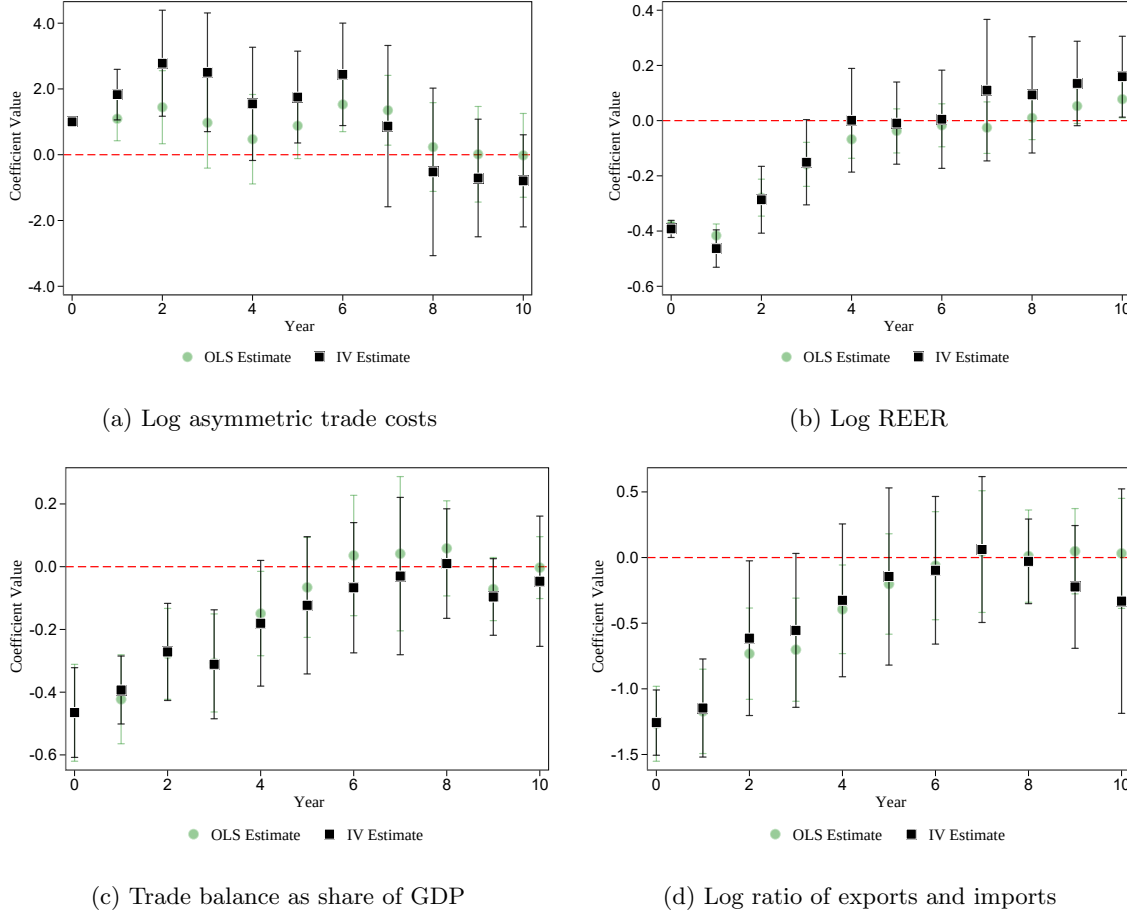
Fact 4: Asymmetric trade costs significantly compress the trade balance and trade ratio. Panels (c) and (d) show that a 1% increase in asymmetric trade costs contracts the export-to-import ratio by 1.3% and reduces the trade balance-to-GDP ratio by 0.45 percentage points, nearly 27% of the standard deviation of its annual change. The trade balance deterioration is consistent with the intertemporal mechanism formalized in Proposition 3: asymmetric costs distort the effective price of international lending, and the resulting real depreciation is insufficient to offset this distortion.

Fact 5: Current account adjustments are driven by shifts in domestic absorption rather than output contraction. A natural question is whether the trade balance deterioration documented in Fact 4 reflects an output contraction or a reallocation of domestic absorption. Figure 9 examines the current account, savings, investment, and real GDP responses. A 1% increase in asymmetric trade costs reduces the current account-to-GDP ratio by 0.48 percentage points on impact, nearly 25% of its annual standard deviation, driven primarily by a surge in investment.¹² Strikingly, real GDP remains unaffected, indicating that the shock does not operate through a supply-side contraction.

The mechanism operates through intertemporal prices: by temporarily raising the relative cost of exporting, asymmetric shocks increase the price of international lending, since cross-border resource transfers ultimately occur through net exports (Ford et al. 2017; Alessandria, Bai, et al. 2025). Faced with a higher cost of transferring resources abroad,

¹²Approximately half of the investment increase in the initial years is due to inventory accumulation (Figure B.3).

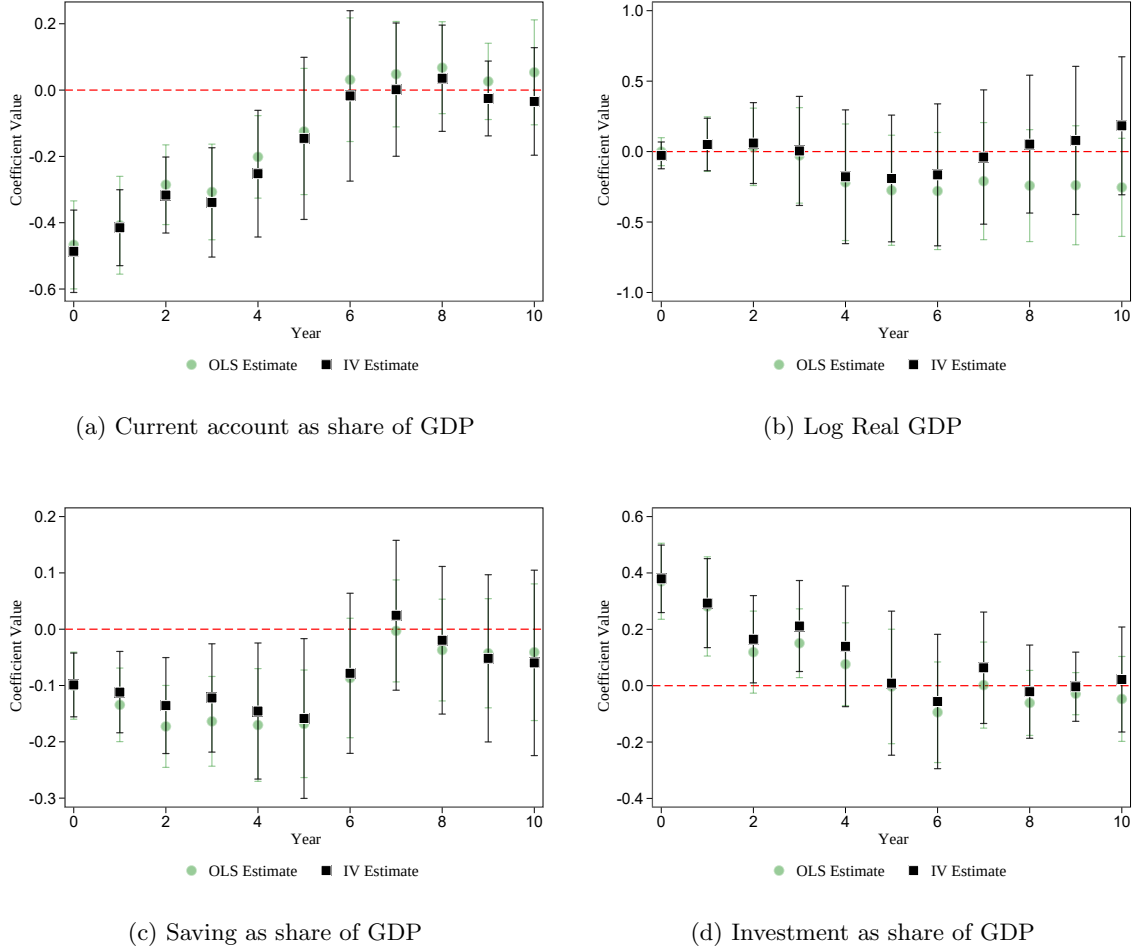
Figure 8. Asymmetric Trade Cost Shocks



Note: IV estimates in square-black, OLS in circle-green. Controls: log terms of trade, log REER, log relative expenditures, four-year lags of all controls and dependent variable. Log values for asymmetric costs, REER, and export-import ratio. 90% confidence intervals; standard errors clustered by country.

domestic agents pull forward expenditure, generating a current account deficit without a corresponding output decline, consistent with our theoretical results.

Figure 9. Asymmetric Trade Cost Shocks and Current Account



Note: IV estimates in square-black, OLS in circle-green. Controls: log terms of trade, log REER, log relative expenditures, four-year lags of all controls and dependent variable. Real GDP in log values. 90% confidence intervals; standard errors clustered by country.

7 Discussion: Tariffs, Common, and Asymmetric Trade Costs

Having established the distinct macroeconomic effects of common versus asymmetric trade cost shocks, we now examine how tariff policy maps into these components. The answer determines whether tariff policy can influence the trade balance or merely acts as a drag on total trade volumes.

We estimate the following specification in yearly changes:

$$\Delta \ln(\tau_{i,t}) = \beta_1 \Delta T_{i,t} + \beta_2 (I_{\Delta T_{i,t} > 0} \times \Delta T_{i,t}) + \beta_3 I_{\Delta T_{i,t} > 0} + \Gamma' \Delta \mathbf{X}_{i,t} + \alpha_t + \epsilon_{i,t} \quad (33)$$

where $\Delta T_{i,t}$ denotes the change in effective tariffs. The control vector $\mathbf{X}_{i,t}$ includes the terms of trade, the REER, and relative expenditure shares, alongside four lags of the dependent variable and the tariff level. The coefficient β_1 captures the baseline elasticity of trade costs to tariff movements, while β_2 identifies the differential response to tariff increases, allowing us to test for asymmetries in protectionist episodes.

Panel 1 of Table 1 shows the results of estimating equation (33) with import costs as the dependent variable. The estimates imply near-unitary pass-through: a 1 percentage point change in effective tariffs is associated with a 1.03% change in import costs, confirming that our measure captures tariff movements after controlling for global shocks and macroeconomic covariates.

Panel 2 shows the results of estimating equation (33) with asymmetric trade costs as the dependent variable. The results reveal a more striking pattern: when tariffs increase, the asymmetric cost measure shows no significant movement, implying that export costs rise in tandem with import costs during protectionist episodes. Tariff increases therefore function as a broad tax on international integration rather than a targeted distortion of the trade balance: the common cost effect documented in Fact 1 dominates, explaining why protectionist policies often fail to improve the external position. For tariff decreases, however, asymmetric trade costs decline significantly, suggesting that liberalization episodes are more likely to be unilateral in nature.

The differential response between tariff increases and decreases points toward foreign retaliation as the operative mechanism. Two observations support this conclusion. First, under our theoretical framework, export and import costs are separately identified, and tariff changes would affect estimated export costs only if there were a structural relationship in which tariffs raise the cost of exporting relative to selling domestically. If such a relationship existed and export costs rose due to production function differences, one would expect a symmetric response to all tariff changes, not only to increases. The fact that the coefficient sign and statistical significance change depending on the direction of tariff movements suggests a mechanism other than structural economic linkages. Foreign retaliation provides a natural explanation, though a more granular exploration of retaliatory dynamics

Table 1. Tariffs and Trade Costs

Panel 1: Import costs			
	Δ Import costs	Δ Import costs	Δ Import costs
Δ Tariffs	2.917*** (0.300)	2.897*** (0.313)	1.034*** (0.151)
N	1037	1037	649
R^2	0.084	0.111	0.941
Panel 2: Asymmetric trade costs			
	Δ Exp-Imp cost	Δ Exp-Imp cost	Δ Exp-Imp cost
Δ Tariffs $\times I_{\Delta \text{Tariffs} \leq 0}$	-5.020*** (0.716)	-5.375*** (0.733)	-1.837*** (0.360)
Δ Tariffs $\times I_{\Delta \text{Tariffs} > 0}$	-1.643 (1.424)	-2.137 (1.433)	0.923 (0.518)
N	1037	1037	649
R^2	0.052	0.085	0.942
Year FE	-	✓	✓
Controls	-	-	✓

Note: Estimates of equation (33). Panel 1: import cost changes. Panel 2: export-import cost ratio changes. Controls: log terms of trade, log REER, log relative expenditures, four-year lags. Standard errors clustered by country.

remains an avenue for future research beyond the scope of this paper. The identification method developed here provides the tools necessary to pursue such an analysis.

8 Conclusion

This paper shows, both empirically and theoretically, that distinguishing between common and asymmetric trade cost shocks is key to understanding their macroeconomic consequences. The long-standing disconnect between theory and evidence on the effects of trade costs on external balances is a measurement problem: standard indices conflate symmetric cost movements, which are neutral for external balances, with asymmetric ones, which are not. We develop a methodology to separately identify these components and show that asymmetric shocks produce significant and persistent current account deteriorations driven

by increased absorption rather than output contraction, even though both types of shocks are temporary. Tariff increases predominantly generate common rather than asymmetric cost movements, providing a new explanation for why protectionism often fails to improve the trade balance.

Our estimated trade cost series open several avenues for future work. They deliver key conditional moments to estimate and discipline quantitative multi-country models, which could enable counterfactual exercises quantifying the relevance of asymmetric trade frictions relative to other sources of macroeconomic fluctuations. The methodology also provides the tools to study retaliatory dynamics more granularly, though the extent to which retaliation fully explains the mapping from tariffs to common costs remains an open question. Finally, our finding that initial trade openness amplifies the macroeconomic effects of asymmetric shocks raises the possibility that the consequences of trade policy fragmentation may fall disproportionately on more open economies, a hypothesis that warrants further investigation.

References

- Alessandria, George, Yan Bai, and Soo Kyung Woo (2025). “Rising Current Account Dispersion: Financial or Trade Integration?” In.
- Alessandria, George and Horag Choi (2021). “The dynamics of the US trade balance and real exchange rate: The J curve and trade costs?” In: *Journal of International Economics* 132, p. 103511.
- Alessandria, George, Horag Choi, and Dan Lu (2017). “Trade integration and the trade balance in China.” In: *IMF Economic Review* 65, pp. 633–674.
- Anderson, James E and Eric Van Wincoop (2004). “Trade costs.” In: *Journal of Economic literature* 42.3, pp. 691–751.
- Anderson, James E. and Eric van Wincoop (2003). “Gravity with Gravitas: A Solution to the Border Puzzle.” In: *American Economic Review* 93.1, pp. 170–192.
- Backus, David, Patrick J Kehoe, and Finn Kydland (1993). *International business cycles: theory and evidence*.
- Barattieri, Alessandro (2022). “Asymmetric Trade Liberalizations and Current Account Dynamics.” In: *Canadian Journal of Economics* 55.2, pp. 905–936.
- Barattieri, Alessandro, Matteo Cacciatore, and Fabio Ghironi (2021). “Macroeconomics of Trade Tariffs: A Developing Country Perspective.” In: *Journal of International Economics* 129, p. 103417.
- Baxter, Marianne and Mario J. Crucini (1995). “Business Cycles and the Asset Structure of Foreign Trade.” In: *International Economic Review* 36.4, pp. 821–854.
- Bayoumi, Tamim, Jaewoo Lee, and Sarma Jayanthi (2006). “New rates from new weights.” In: *IMF Staff Papers* 53.2, pp. 272–305.
- Bems, Rudolfs, Robert C Johnson, and Kei-Mu Yi (2013). “The great trade collapse.” In: *Annu. Rev. Econ.* 5.1, pp. 375–400.
- Boer, Lukas and Malte Rieth (2024). *The macroeconomic consequences of import tariffs and trade policy uncertainty*. International Monetary Fund.
- Boz, Ms Emine, MissNan Li, and Hongrui Zhang (2019). *Effective trade costs and the current account: An empirical analysis*. International Monetary Fund.
- Chinn, Menzie D. and Eswar S. Prasad (2003). “Medium-Term Determinants of Current Accounts in Industrial and Developing Countries: An Empirical Exploration.” In: *Journal of International Economics* 59.1, pp. 47–76.
- Costinot, Arnaud and Iván Werning (2025). *How tariffs affect trade deficits*. Tech. rep. National Bureau of Economic Research.

- Darvas, Zsolt M (2021). *Timely measurement of real effective exchange rates*. Tech. rep. Bruegel working paper.
- Dekle, Robert, Jonathan Eaton, and Samuel Kortum (2007). “Unbalanced Trade.” In: *American Economic Review* 97.2, pp. 351–355.
- (2008). “Global Rebalancing with Gravity: Measuring the Burden of Adjustment.” In: *IMF Staff Papers* 55.3, pp. 511–540.
- Dube, Arindrajit, Daniele Girardi, Oscar Jorda, and Alan M Taylor (2023). *A local projections approach to difference-in-differences event studies*. Tech. rep. National Bureau of Economic Research Cambridge, Massachusetts.
- Eaton, Jonathan and Samuel Kortum (2002). “Technology, geography, and trade.” In: *Econometrica* 70.5, pp. 1741–1779.
- Eaton, Jonathan, Samuel Kortum, Brent Neiman, and John Romalis (2016). “Trade and the global recession.” In: *American Economic Review* 106.11, pp. 3401–3438.
- Erceg, Christopher J., Andrea Prestipino, and Andrea Raffo (2018). *The Macroeconomic Effects of Trade Policy*. International Finance Discussion Papers 1242. Board of Governors of the Federal Reserve System.
- Estefania-Flores, Julia, Davide Furceri, Swarnali A Hannan, Jonathan D Ostry, and Andrew K Rose (2024). “A measurement of aggregate trade restrictions and their economic effects.” In: *The World Bank Economic Review*.
- (2025). “A measurement of aggregate trade restrictions and their economic effects.” In: *The World Bank Economic Review* 39.3, pp. 684–710.
- Fajgelbaum, Pablo D, Pinelopi K Goldberg, Patrick J Kennedy, and Amit K Khandelwal (2020). “The return to protectionism.” In: *The quarterly journal of economics* 135.1, pp. 1–55.
- Fitzgerald, Doireann (2012). “Trade costs, asset market frictions, and risk sharing.” In: *American Economic Review* 102.6, pp. 2700–2733.
- Fitzgerald, Doireann, Yaniv Yedid Levi, and Stefanie Haller (2024). “Can Sticky Quantities Explain Export Insensitivity to Exchange Rates?” In: *IMF Economic Review*, pp. 1–25.
- Ford, Nicholas and Charles Yuji Horioka (2017). “The ‘real’ explanation of the Feldstein–Horioka puzzle.” In: *Applied Economics Letters* 24.2, pp. 95–97.
- Furceri, Davide, Swarnali A Hannan, Jonathan D Ostry, and Andrew K Rose (2022). “The macroeconomy after tariffs.” In: *The World Bank Economic Review* 36.2, pp. 361–381.
- Gopinath, Gita, Pierre-Olivier Gourinchas, Andrea F Presbitero, and Petia Topalova (2025). “Changing Global Linkages: Bridging Geopolitical Fragments.” In: *AEA Papers and Proceedings*. Vol. 115. American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 605–610.

- Gourinchas, Pierre-Olivier and Hélène Rey (2007). “From World Banker to World Venture Capitalist: US External Adjustment and the Exorbitant Privilege.” In: *G7 Current Account Imbalances: Sustainability and Adjustment*. Ed. by Richard Clarida. University of Chicago Press, pp. 11–66.
- Head, Keith and Thierry Mayer (2014). “Gravity Equations: Workhorse, Toolkit, and Cookbook.” In: *Handbook of International Economics*. Ed. by Gita Gopinath, Elhanan Helpman, and Kenneth Rogoff. Vol. 4. Elsevier, pp. 131–195.
- Head, Keith and John Ries (2001). “Increasing returns versus national product differentiation as an explanation for the pattern of US–Canada trade.” In: *American Economic Review* 91.4, pp. 858–876.
- Herman, Peter (2023). *An Introduction to the GEPPML Gravity Estimation: A Practical Guide*. Tech. rep. USITC Economics Working Paper.
- Irwin, Douglas A (2000). *Tariffs and growth in late nineteenth century America*.
- Itskhoki, Oleg and Dmitry Mukhin (2025). “Sanctions and the exchange rate.” In: *Review of Economic Studies*, rdaf085.
- Jeanne, Olivier and Jeongwon Son (2024). “To What Extent Are Tariffs Offset by Exchange Rates?” In: *Journal of International Money and Finance*. DOI: [10.1016/j.jimonfin.2024.103015](https://doi.org/10.1016/j.jimonfin.2024.103015).
- Jordà, Òscar and Alan M Taylor (2025). “Local projections.” In: *Journal of Economic Literature* 63.1, pp. 59–110.
- Kohn, David, Fernando Leibovici, and Michal Szkup (2016). “Financial frictions and new exporter dynamics.” In: *International economic review* 57.2, pp. 453–486.
- Lerner, Abba P (1936). “The symmetry between import and export taxes.” In: *Economica* 3.11, pp. 306–313.
- (1944). *The Economics of Control: Principles of Welfare Economics*. New York: Macmillan.
- Mac Mullen, Marcos and Soo Kyung Woo (2023). *Real exchange rate and net trade dynamics: financial and trade shocks*. Tech. rep. University of Rochester working paper.[5, 17, 19].
- Novy, Dennis (2013). “Gravity Redux: Measuring International Trade Costs with Panel Data.” In: *Economic Inquiry* 51.1, pp. 101–121.
- Obstfeld, Maurice and Kenneth Rogoff (1995). “The Intertemporal Approach to the Current Account.” In: *Handbook of International Economics*. Ed. by Gene M. Grossman and Kenneth Rogoff. Vol. 3. Elsevier, pp. 1731–1799.
- (2005). “Global Current Account Imbalances and Exchange Rate Adjustments.” In: *Brookings Papers on Economic Activity* 36.1, pp. 67–146.

- Piger, Jeremy and Thomas Stockwell (2025). “Differences from differencing: Should local projections with observed shocks be estimated in levels or differences?” In: *Journal of Applied Econometrics* 40.7, pp. 759–787.
- Reyes-Heroles, Ricardo et al. (2016). “The role of trade costs in the surge of trade imbalances.” In: *Princeton University, mimeograph*.
- Robinson, Joan (1937). “The Foreign Exchanges.” In: *Economica* 4.13, pp. 1–20.
- Rogoff, Kenneth S and Maurice Obstfeld (2000). “The six major puzzles in international macroeconomics: Is there a common cause?” In.
- Schmitt-Grohé, Stephanie and Martín Uribe (2025). *Transitory and Permanent Import Tariff Shocks in the United States: An Empirical Investigation*. Tech. rep. National Bureau of Economic Research.
- Simonovska, Ina and Michael E Waugh (2014). “The elasticity of trade: Estimates and evidence.” In: *Journal of international Economics* 92.1, pp. 34–50.
- Waugh, Michael E (2010). “International trade and income differences.” In: *American Economic Review* 100.5, pp. 2093–2124.
- World Bank (2011). *International Comparison Program: Methodological Handbook*. Tech. rep. Washington, DC: World Bank.
- Yotov, Yoto V (2022). “On the role of domestic trade flows for estimating the gravity model of trade.” In: *Contemporary Economic Policy* 40.3, pp. 526–540.

A Proofs

A.1 Proof of Proposition 1

Proof. From equation (10), the estimated fixed effects satisfy:

$$\hat{\gamma}_i = (1 - \sigma) (\ln p_{ii} + \ln \tau_i^X) \quad (34)$$

$$\hat{\delta}_j = (1 - \sigma) (-\ln p_{jj} + \ln \tau_j^M) \quad (35)$$

$$\hat{\epsilon}_{ij} = (1 - \sigma) \ln \tau_{ij}^B \quad (36)$$

Part 1 (Bilateral costs). From the third equation, $\hat{\tau}_{ij}^B = \frac{\hat{\epsilon}_{ij}}{(1-\sigma)}$ directly, regardless of macro cost heterogeneity. $\square$

Part 2 (Country-level directional costs). Combining the exporter and importer fixed effects for the *same* country i :

$$\frac{\hat{\gamma}_i + \hat{\delta}_i}{1 - \sigma} = \ln \tau_i^X + \ln \tau_i^M \quad (37)$$

The estimated total cost from i to j is therefore:

$$\ln \hat{\tau}_{ij} = \ln \tau_{ij}^B + \ln \tau_i^X + \ln \tau_i^M \quad (38)$$

The true total cost under Assumption 4 is:

$$\ln \tau_{ij} = \ln \tau_{ij}^B + \ln \tau_i^X + \ln \tau_j^M \quad (39)$$

The bias in logs is:

$$\ln \hat{\tau}_{ij} - \ln \tau_{ij} = \ln \tau_i^M - \ln \tau_j^M \quad (40)$$

Sufficiency ($\Leftarrow$): If $\tau_j^M = \alpha^M$ for all j , then equation (40) equals zero for all pairs (i, j) , so $\hat{\tau}_{ij} = \tau_{ij}$. The analogous argument holds if $\tau_i^X = \alpha^X$ for all i .

Necessity ($\Rightarrow$): If $\hat{\tau}_{ij} = \tau_{ij}$ for all (i, j) , then $\ln \tau_i^M = \ln \tau_j^M$ for all i, j , implying $\tau_j^M = \alpha^M$

for some constant α^M . Symmetrically, if the estimation normalizes $\tau_j^M = 1$ rather than $\tau_i^X = 1$, the bias becomes $\ln \tau_i^X - \ln \tau_j^X$, which vanishes if and only if $\tau_i^X = \alpha^X$ for all i .

□

□

A.2 Proof of Proposition 2

Proof. We denote country d 's bilateral expenditure share on goods from country j as $s_{d,j} \equiv (1 - \varphi) (\tau_{d,j} p_{d,d} / P_j)^{1-\sigma}$ and $s_{\text{ROW},d} \equiv \sum_{j \neq d} s_{j,d}$. We define the aggregate ROW price-expenditure terms as $P_f^\sigma C_f \equiv \sum_{j \neq d} P_j^\sigma C_j$ and $p_{f,f}^{1-\sigma} \equiv \sum_{j \neq d} p_{j,j}^{1-\sigma}$. Throughout, $P_t^d \equiv P_{d,t}$ and $P_t^f \equiv P_{f,t}$.

Parts (i) and (ii): Export Mapping.

Country d 's total exports in the multi-country model:

$$\text{EXP}_d = (1 - \varphi) p_{d,d}^{1-\sigma} (\tau_d^X)^{-(\sigma-1)} \sum_{j \neq d} \frac{P_j^\sigma C_j}{(\tau_j^M \cdot \tau_{d,j}^B)^{\sigma-1}}$$

In the two-country model:

$$\text{EXP}_d^{2c} = (1 - \varphi) p_{d,d}^{1-\sigma} (\tilde{\tau}_d^X)^{-(\sigma-1)} \frac{P_f^\sigma C_f}{(\tilde{\tau}_f^M)^{\sigma-1}}$$

Matching $\text{EXP}_d^{2c} = \text{EXP}_d$ and imposing $\tilde{\tau}_d^X = \tau_d^X$:

$$\frac{P_f^\sigma C_f}{(\tilde{\tau}_f^M)^{\sigma-1}} = \sum_{j \neq d} \frac{P_j^\sigma C_j}{(\tau_j^M \cdot \tau_{d,j}^B)^{\sigma-1}}$$

From the expenditure share $s_{d,j} = (1 - \varphi) (\tau_{d,j} p_{d,d} / P_j)^{1-\sigma}$:

$$\frac{P_j^\sigma C_j}{(\tau_j^M \cdot \tau_{d,j}^B)^{\sigma-1}} = \frac{s_{d,j} P_j C_j}{(1 - \varphi) p_{d,d}^{1-\sigma}} \cdot (\tau_d^X)^{\sigma-1}$$

Substituting and using $\text{EXP}_d = \sum_{j \neq d} s_{d,j} P_j C_j$:

$$(\tilde{\tau}_f^M)^{\sigma-1} = \sum_{j \neq d} \frac{s_{d,j} P_j C_j}{\text{EXP}_d} \cdot (\tau_j^M \cdot \tau_{d,j}^B)^{\sigma-1}$$

Part (iii): Import Mapping.

Country d 's total imports in the multi-country model:

$$\text{IMP}_d = (1 - \varphi) P_d^\sigma C_d (\tau_d^M)^{-(\sigma-1)} \sum_{j \neq d} \frac{p_{j,j}^{1-\sigma}}{(\tau_j^X \cdot \tau_{j,d}^B)^{\sigma-1}}$$

In the two-country model:

$$\text{IMP}_d^{2c} = (1 - \varphi) P_d^\sigma C_d (\tilde{\tau}_d^M)^{-(\sigma-1)} \frac{p_{f,f}^{1-\sigma}}{(\tilde{\tau}_f^X)^{\sigma-1}}$$

Matching $\text{IMP}_d^{2c} = \text{IMP}_d$ and imposing $\tilde{\tau}_d^M = \tau_d^M$:

$$\frac{p_{f,f}^{1-\sigma}}{(\tilde{\tau}_f^X)^{\sigma-1}} = \sum_{j \neq d} \frac{p_{j,j}^{1-\sigma}}{(\tau_j^X \cdot \tau_{j,d}^B)^{\sigma-1}}$$

From the expenditure share $s_{j,d} = (1 - \varphi) (\tau_{j,d} p_{j,j} / P_d)^{1-\sigma}$:

$$\frac{p_{j,j}^{1-\sigma}}{(\tau_j^X \cdot \tau_{j,d}^B)^{\sigma-1}} = \frac{s_{j,d} P_d^{1-\sigma}}{1 - \varphi} \cdot (\tau_d^M)^{\sigma-1}$$

Substituting and using $p_{f,f}^{1-\sigma} = \sum_{j \neq d} p_{j,j}^{1-\sigma}$ and $s_{\text{ROW},d} = \sum_{j \neq d} s_{j,d}$:

$$(\tilde{\tau}_f^X)^{\sigma-1} = \sum_{j \neq d} \frac{s_{j,d}}{s_{\text{ROW},d}} \cdot (\tau_j^X \cdot \tau_{j,d}^B)^{\sigma-1}$$

□

A.3 Proof of Proposition 3

Log-Linearization

Throughout this proof, country labels d and f denote domestic and rest-of-world respectively, with $P_t^d \equiv P_{d,t}$ and $P_t^f \equiv P_{f,t}$. Hat notation: $\hat{x} \equiv d \ln x$; $\tilde{b} \equiv b/(\bar{P}\bar{C})$ (level deviation since $\bar{b} = 0$); $z \equiv \hat{q}_1 - \hat{q}_2$; $\Delta \equiv \hat{\tau}_{fd,1} - \hat{\tau}_{df,1}$.

From the model equilibrium conditions, log-linearization gives:

$$\hat{P}_t^d = \lambda(\hat{q}_t + \hat{\tau}_{fd,t}), \quad \hat{P}_t^f = (1 - \lambda)\hat{q}_t + \lambda\hat{\tau}_{df,t}$$

In period 2: $\hat{P}_2^d = \lambda\hat{q}_2$, $\hat{P}_2^f = (1 - \lambda)\hat{q}_2$. From Step 4: $\hat{k}^f = \hat{k}^d + z/\eta$.

Linearizing the 8 equilibrium equations:

(La):

$$\gamma(\hat{C}_2^d - \hat{C}_1^d) + (\hat{P}_2^d - \hat{P}_1^d) = -\eta\hat{k}^d \quad (\text{La})$$

(Lb):

$$\gamma(\hat{C}_2^f - \hat{C}_1^f) + (\hat{P}_2^f - \hat{P}_1^f) = -\eta\hat{k}^d \quad (\text{Lb})$$

(Lc): Linearize $y_1 = P_1^d C_1^d + k^d + b$, divide by y_1 . Use $\bar{P}\bar{C}/y_1 = \xi$, $\bar{k}/y_1 = \omega$:

$$\hat{C}_1^d = -\hat{P}_1^d - \frac{\omega}{\xi}\hat{k}^d - \tilde{b} \quad (\text{Lc})$$

(Ld): Linearize $f(k^d) + Rb = P_2^d C_2^d$. Since $\bar{b} = 0$: $d(Rb) = R db$. Divide by $\bar{P}\bar{C}$:

$$\hat{C}_2^d = \phi\hat{k}^d + R\tilde{b} - \hat{P}_2^d \quad (\text{Ld})$$

(Le): Linearize $q_1 y_1 + b = P_1^f C_1^f + q_1 k^f$. Divide by $\bar{P}\bar{C}$:

$$\hat{C}_1^f = \hat{q}_1 + \tilde{b} - \hat{P}_1^f - \frac{\omega}{\xi}\hat{k}^d - \frac{\omega}{\xi\eta}z \quad (\text{Le})$$

(Lf): Linearize $q_2 f(k^f) = P_2^f C_2^f + Rb$. Divide by $\bar{P}\bar{C}$:

$$\hat{C}_2^f = \hat{q}_2 + \phi \hat{k}^d + \frac{\phi}{\eta} z - R\tilde{b} - \hat{P}_2^f \quad (\text{Lf})$$

(Lg): Steady-state shares of good f : $(1-\lambda)\xi$ domestic, $\lambda\xi$ imports, ω investment.

$$0 = -2\sigma\lambda(1-\lambda)\xi\hat{q}_1 + (1-\lambda)\xi\hat{C}_1^f + \lambda\xi\hat{C}_1^d + T_g + \omega\hat{k}^d + \frac{\omega}{\eta}z \quad (\text{Lg})$$

where $T_g = \sigma\lambda(1-\lambda)\xi\hat{\tau}_{fd,1} + \lambda\xi[1-\sigma(1-\lambda)]\hat{\tau}_{df,1}$.

(Lh): Steady-state shares of good f in period 2: $(1-\lambda)$ domestic, λ exports.

$$\phi(\hat{k}^d + z/\eta) = -2\sigma\lambda(1-\lambda)\hat{q}_2 + (1-\lambda)\hat{C}_2^f + \lambda\hat{C}_2^d \quad (\text{Lh})$$

These are **8 linear equations in 8 unknowns** ($\hat{C}_1^d, \hat{C}_2^d, \hat{C}_1^f, \hat{C}_2^f, \hat{k}^d, \tilde{b}, \hat{q}_1, \hat{q}_2$).

Solving for the Current Account

We use the budget constraints (Lc)–(Lf) to substitute $\hat{C}_t^i$ into the Euler and goods market equations (La), (Lb), (Lg), (Lh).

Step 1: Substitute (Lc), (Le) into (Lg). The $\hat{k}^d$ terms cancel. After substituting the $\hat{P}$ identities and collecting terms:

$$M\hat{q}_1 + \frac{\kappa-1}{2}\Delta - \frac{\lambda\omega}{\xi\eta}z = (1-2\lambda)\tilde{b} \quad (\text{G-I})$$

Step 2: Substitute (Ld), (Lf) into (Lh). The $\hat{k}^d$ terms cancel.

$$M\hat{q}_2 + \frac{\lambda R\omega}{\xi\eta}z = -(1-2\lambda)R\tilde{b} \quad (\text{G-II})$$

Step 3: Substitute (Lc), (Ld) into (La). Compute $\hat{C}_2^d - \hat{C}_1^d = \omega\psi\hat{k}^d + \psi\xi\tilde{b} + \hat{P}_1^d - \hat{P}_2^d$. Substituting into (La):

$$(\gamma\omega\psi + \eta)\hat{k}^d + \gamma\psi\xi\tilde{b} + (\gamma-1)(\hat{P}_1^d - \hat{P}_2^d) = 0 \quad (\text{E-I})$$

Step 4: Substitute (Le), (Lf) into (Lb). Compute $\hat{C}_2^f - \hat{C}_1^f = -z + \omega\psi\hat{k}^d + \omega\psi z/\eta - \psi\xi\tilde{b} + \hat{P}_1^f - \hat{P}_2^f$. Substituting into (Lb):

$$(\gamma\omega\psi + \eta)\hat{k}^d - \gamma\psi\xi\tilde{b} + \gamma(-z + \omega\psi z/\eta) + (\gamma-1)(\hat{P}_1^f - \hat{P}_2^f) = 0 \quad (\text{E-II})$$

Step 5: Subtract (E-II) from (E-I) to eliminate $\hat{k}^d$. The $(\gamma\omega\psi + \eta)\hat{k}^d$ terms cancel. Substituting $\hat{P}_1^d - \hat{P}_2^d = \lambda(z + \hat{\tau}_{fd,1})$ and $\hat{P}_1^f - \hat{P}_2^f = (1-\lambda)z + \lambda\hat{\tau}_{df,1}$:

$$2\gamma\psi\xi\tilde{b} + \left[(\gamma-1)(2\lambda-1) + \gamma - \frac{\gamma\omega\psi}{\eta} \right] z + (\gamma-1)\lambda\Delta = 0 \quad (\text{E-diff})$$

Step 6: Combine (G-I) and (G-II) to eliminate $\hat{q}_2$. From (G-II): $M\hat{q}_2 = -(1-2\lambda)R\tilde{b} - \lambda R\omega z/(\xi\eta)$. Substituting $\hat{q}_1 = z + \hat{q}_2$ into (G-I):

$$\left[M - \frac{\lambda\omega\psi}{\eta} \right] z + \frac{\kappa-1}{2} \Delta = (1-2\lambda)\psi\xi\tilde{b} \quad (\text{G-comb})$$

Step 7: Solve the 2×2 system (E-diff), (G-comb) for $(\tilde{b}, z)$. From (E-diff):

$$\tilde{b} = \frac{-\left[(\gamma-1)(2\lambda-1) + \gamma - \gamma\omega\psi/\eta \right] z - (\gamma-1)\lambda\Delta}{2\gamma\psi\xi}$$

Substituting into (G-comb) and solving for z yields $z = -(N_\Delta/D_z)\Delta$, where D_z and N_Δ collect the relevant coefficients. Then:

$$\tilde{b} = \Phi\Delta, \quad \Phi = \frac{\left[M - \frac{\lambda\omega\psi}{\eta} \right] (N_\Delta/D_z) - \frac{\kappa-1}{2}}{(1-2\lambda)\psi\xi} \quad (41)$$

A.4 Proof of Proposition 5

Proof. The expenditure shares in the two-country model satisfy:

$$s_{dft} = (1 - \varphi) \left(\frac{p_{dft}}{P_{ft}} \right)^{1-\sigma}, \quad s_{ddt} = \varphi \left(\frac{p_{ddt}}{P_{dt}} \right)^{1-\sigma}$$

Using $p_{dft} = \tau_{dft} p_{ddt}$, the ratio of expenditure shares is:

$$\frac{s_{dft}}{s_{ddt}} = \frac{1 - \varphi}{\varphi} \cdot \frac{(\tau_{dft} p_{ddt} / P_{ft})^{1-\sigma}}{(p_{ddt} / P_{dt})^{1-\sigma}} = \frac{1 - \varphi}{\varphi} (\tau_{dft})^{1-\sigma} \left(\frac{P_{dt}}{P_{ft}} \right)^{1-\sigma}$$

Defining $\mathcal{X}_{dt} \equiv \left(\frac{1-\varphi}{\varphi} \right)^{\frac{1}{1-\sigma}} \tau_{dft}$ and raising to the power $\frac{1}{1-\sigma}$:

$$\mathcal{X}_{dt} = \left(\frac{s_{dft}}{s_{ddt}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{ft}}{P_{dt}} \right)$$

By an analogous argument for imports, using $s_{fdt} = (1 - \varphi) (\tau_{fdt} p_{fft} / P_{dt})^{1-\sigma}$ and $s_{fft} = \varphi (p_{fft} / P_{ft})^{1-\sigma}$, and defining $\mathcal{M}_{dt} \equiv \left(\frac{\varphi}{1-\varphi} \right)^{\frac{1}{1-\sigma}} \tau_{fdt}$:

$$\mathcal{M}_{dt} = \left(\frac{s_{fdt}}{s_{fft}} \right)^{\frac{1}{1-\sigma}} \left(\frac{P_{dt}}{P_{ft}} \right)$$

□

A.5 Proof of Proposition 6 and Corollary 7

Proof. Let $y_1 = \mathcal{X}_{dt}$ and $y_2 = \mathcal{M}_{dt}$ for notational simplicity.

($\Rightarrow$) Forward Direction: Conditions (i)–(ii) imply decomposition exists.

Step 1: Construct the inverse mapping. Define $T^{-1} : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+^2$ by:

$$\tau_{d,t}^C = y_1^{1/2} y_2^{1/2} \tag{42}$$

$$\tau_{d,t}^A = \left(\frac{y_1}{y_2} \right)^\mu \tag{43}$$

where $\mu > 0$ is a free parameter.

Step 2: Verify invertibility. The Jacobian determinant is:

$$\det(J^{-1}) = -\mu y_1^{\mu-1/2} y_2^{-\mu-1/2} \neq 0 \quad (44)$$

since $\mu > 0$. Non-collinearity of y_1 and y_2 (condition (ii)) guarantees this holds everywhere on $\mathbb{R}_+^2$.

Step 3: Apply the Inverse Function Theorem. Since $\det(J^{-1}) \neq 0$ and the component functions are $\mathcal{C}^2$ on $\mathbb{R}_+^2$, there exists a local inverse $T \in \mathcal{C}^2$ with $y_1 = f(\tau^C, \tau^A)$ and $y_2 = g(\tau^C, \tau^A)$.

Step 4: Derive functional forms and verify elasticity constraints. Solving the inverse system:

$$y_1 = \tau^C \cdot (\tau^A)^{1/(2\mu)} \equiv f(\tau^C, \tau^A) \quad (45)$$

$$y_2 = \tau^C \cdot (\tau^A)^{-1/(2\mu)} \equiv g(\tau^C, \tau^A) \quad (46)$$

The elasticities satisfy:

$$\varepsilon_{f,\tau^C} = \varepsilon_{g,\tau^C} = 1 \quad (47)$$

$$\varepsilon_{f,\tau^A} = \frac{1}{2\mu} \neq -\frac{1}{2\mu} = \varepsilon_{g,\tau^A} \quad (48)$$

Both constraints from Definitions 1 and 2 are satisfied for any $\mu > 0$.

($\Leftarrow$) Reverse Direction: If decomposition exists, conditions (i)–(ii) must hold.

Condition (i): The elasticity constraints require well-defined partial derivatives of f and g that are preserved under small perturbations. By the Inverse Function Theorem, this requires $f, g \in \mathcal{C}^2(\mathbb{R}_+^2)$.

Condition (ii): Suppose for contradiction that $y_2 = ky_1 + c'$ for constants k, c' . Then:

$$\varepsilon_{g,\tau^A} = \frac{ky_1}{ky_1 + c'} \varepsilon_{f,\tau^A} \quad (49)$$

For $\varepsilon_{f,\tau^A} \neq \varepsilon_{g,\tau^A}$ to hold with a fixed decomposition requires $c' \neq 0$, but then the ratio $\varepsilon_{g,\tau^A}/\varepsilon_{f,\tau^A}$ varies with y_1 across (d, t) , contradicting the existence of a fixed decomposition with constant elasticity difference ε^A . Therefore perfect collinearity is impossible.

Proof of Corollary 7.

Given a valid decomposition (f, g, τ^C, τ^A) and any strictly monotonic $h_1, h_2 \in \mathcal{C}^2$, define $\tilde{\tau}^C = h_1(\tau^C)$ and $\tilde{\tau}^A = h_2(\tau^A)$. Then $\tilde{f}(\tilde{\tau}^C, \tilde{\tau}^A) = f(h_1^{-1}(\tilde{\tau}^C), h_2^{-1}(\tilde{\tau}^A))$ yields another valid decomposition since by the chain rule:

$$\varepsilon_{\tilde{f}, \tilde{\tau}^C} = \varepsilon_{f, \tau^C} \cdot \frac{h'_1(\tau^C)\tau^C}{h_1(\tau^C)} = \varepsilon_{\tilde{g}, \tilde{\tau}^C} \quad (50)$$

and $\text{sgn}(\varepsilon_{\tilde{f}, \tilde{\tau}^A} - \varepsilon_{\tilde{g}, \tilde{\tau}^A}) = \text{sgn}(\varepsilon_{f, \tau^A} - \varepsilon_{g, \tau^A})$ since $h'_2 > 0$. This establishes that valid decompositions are identified up to monotonic transformations, i.e., up to a scaling constant. $\square$

B Appendix: Construction of Alternative Measures

B.1 Appendix: Construction of other measures

Here we explain the procedure we use to construct the other trade measures.

B.1.1 Gravity measure

First step: Estimation. Bilateral trade barriers are estimated using a gravity approach extending Eaton and Kortum (2002, Ecta) to panel data. The gravity equation relates bilateral expenditure shares to trade barriers:

$$\frac{\lambda_{id}}{\lambda_{dd}} = \exp\{-\theta(\ln p_i - \ln p_d + \ln \tau_{id})\} \quad (51)$$

where λ_{id} denotes country j 's expenditure share on country i output, p_i represents output prices, τ_{id} captures bilateral trade barriers, and θ is the trade elasticity parameter. The shares are computed using both merchandise and services expenditure.

The estimation assumes common importer effects in trade barriers. The panel specification becomes:

$$\frac{\lambda_{idt}}{\lambda_{jdt}} = \exp\{\pi_{it} + \pi_{dt} + \delta_{dt} + \delta_{id} + \epsilon_{idt}\} \quad (52)$$

where π_{it} and δ_{dt} are time-varying importer and exporter fixed effects, δ_{id} captures time-invariant bilateral factors. The model is estimated using Poisson maximum likelihood estimation.

Second step: Bilateral trade barriers. From the estimate compute the bilateral trade barriers:

$$-\theta \ln \hat{\tau}_{idt}^{EK} = \pi_{it} - \pi_{dt} + \delta_{dt} + \delta_{id} + \epsilon_{idt}$$

Third step: Aggregation. Second, bilateral barriers are aggregated to country-level import and export barrier indices.

Second, bilateral barriers are aggregated to country-level import and export barrier

indices. Import barriers for country d are constructed as:

$$\ln \hat{\tau}_{dt}^{M,EK} = \sum_{i \neq d} \mu_{idt-1} \ln \hat{\tau}_{idt}^{EK} \quad (53)$$

where $\mu_{idt} = M_{idt} / \sum_{i \neq d} M_{idt}$ represents lagged import weights.

Export barriers for country i are analogously constructed as:

$$\ln \hat{\tau}_{it}^{X,EK} = \sum_{d \neq i} \xi_{idt-1} \ln \hat{\tau}_{idt}^{EK} \quad (54)$$

where $\xi_{idt} = M_{idt} / \sum_{j \neq i} M_{idt}$ represents lagged export weights.

B.1.2 Bilateral H-R Index

First step: Compute Bilateral trade barriers. Compute the bilateral expenditure share for countries d and origin i for each period as explain in the main text. Then set the value for θ and compute:

$$\hat{\tau}_{idt}^{HR} = \left(\frac{\lambda_t^{i,d} \lambda_t^{d,i}}{\lambda^{i,i} \lambda^{d,d}} \right)^{\frac{1}{2(1-\theta)}}$$

Second step: Aggregation. We follow the same procedure previously described to aggregate trade costs into export and import aggregate trade costs.

B.1.3 Large Asymmetric Trade Policy Events

Based on the IMF's Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER) database, we construct a classifier that sorts countries into a control and treatment group. Based on the *Policy Asymmetry Index* outlined in equation (27), we classify an event as a *large asymmetric trade policy event* (LATPE) if the change in policy asymmetry is in the top 25th percentile of the distribution. Below is a list of countries that have experienced such a LATPE, which has persisted for at least five years between 2000 and 2019, and the year in which the event first occurred:

Table B.1. Countries and Years for each LATPE

Country	Year	Country	Year	Country	Year
Afghanistan	2011	Gambia	2005	Russia	2003
Algeria	2007	Greece	2004	Samoa	2003
Angola	2013	Haiti	2009	San Marino	2005
Argentina	2014	Hungary	2003	Senegal	2004
Armenia	2005	Iraq	2005	Seychelles	2005
Aruba	2005	Italy	2005	Slovak Republic	2002
Azerbaijan	2001	Jamaica	2000	Slovenia	2003
Bahrain	2004	Japan	2005	Solomon Islands	2005
Belgium	2010	Kazakhstan	2005	South Korea	2014
Belize	2014	Latvia	2005	Spain	2002
Bolivia	2007	Lesotho	2005	Sri Lanka	2002
Bosnia and Herzegovina	2005	Lithuania	2005	Sudan	2011
Canada	2005	Malta	2007	Tajikistan	2005
Cape Verde	2005	Mozambique	2000	Thailand	2005
China	2006	Norway	2013	Turkey	2005
Cote d'Ivoire	2000	Oman	2003	Uzbekistan	2006
Croatia	2003, 2010	Pakistan	2003	Venezuela	2002
Cyprus	2000, 2014	Panama	2008	Yemen	2005
Czech Republic	2005	Papua New Guinea	2001	Zambia	2004
Ecuador	2004	Poland	2005	El Salvador	2005
Romania	2010				

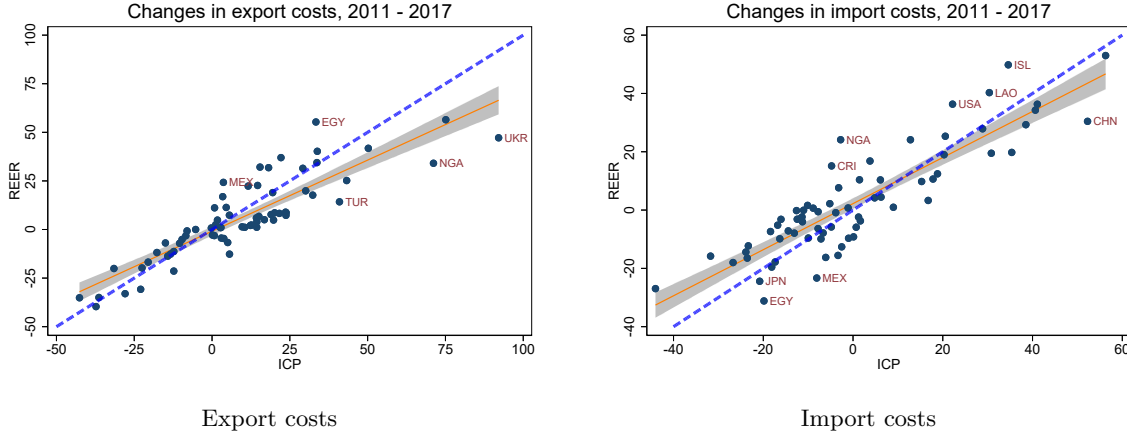
B.2 Appendix: Tables

Table B.2. Instrumental variable: F-statistics

Horizon	F-statistics	Observations
0	62.00	376
1	63.76	362
2	65.69	346
3	62.90	328
4	67.70	305
5	66.10	284
6	64.13	260
7	69.84	241
8	75.52	222
9	70.92	203
10	59.40	179

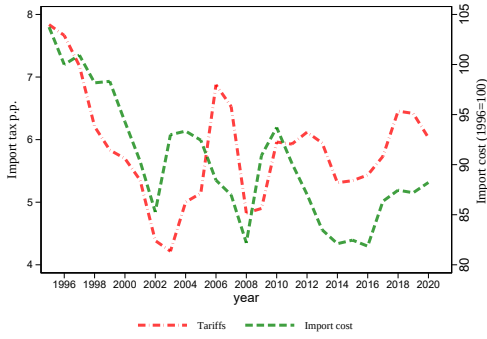
Table B.3. The table shows the Kleibergen-Paap Wald F statistic (column 2), and the number of observation column (3), for the estimation detailed in equation (32)

B.3 Appendix: Figures

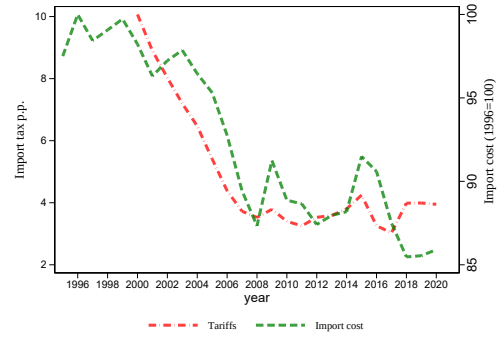
Figure B.1. Trade cost differences: REER vs ICP

Note: This figure shows the change in exports and import costs estimated using two different prices. Y-axis uses trade costs using REER as proxy for prices, while X-axis uses the proxy based on only tradable goods for the ICP database.

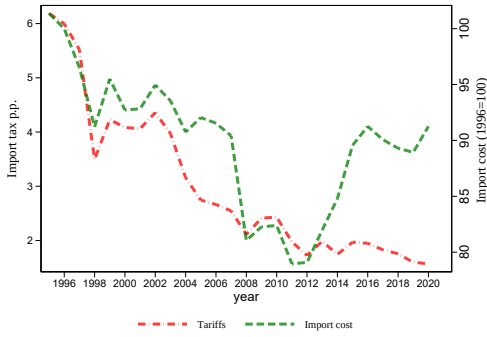
Figure B.2. Effective tariffs vs import costs



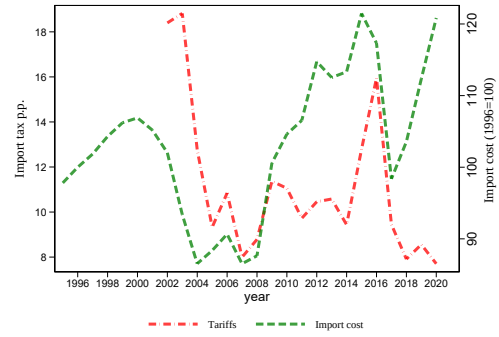
South Africa



Tunisia



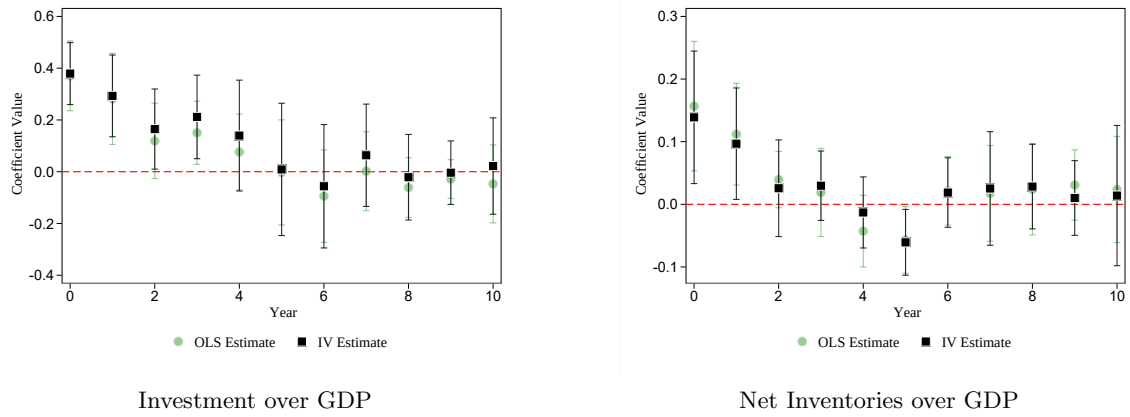
Korea



Egypt

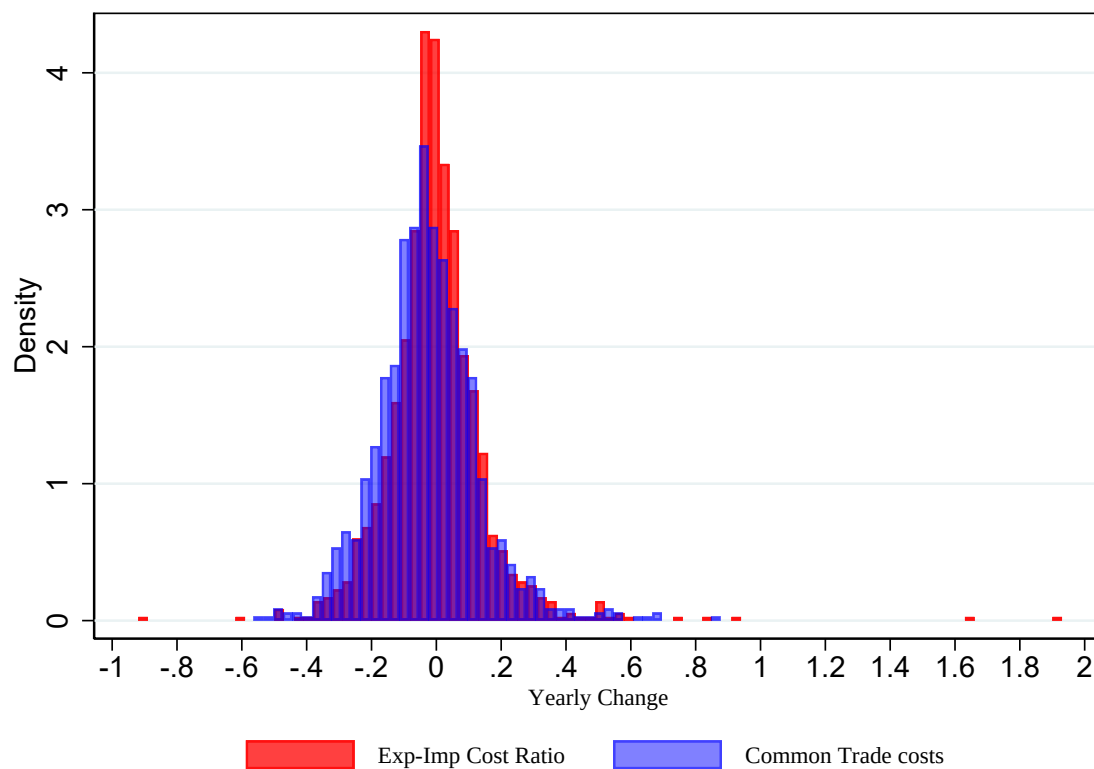
Note: We show the evolution of effective tariffs versus the evolution of import costs for a subset of those countries that went over an cumulative effective tariffs increase of at least 2 p.p. in four years. Effective tariffs are defined as government revenues from imports over total imports. Import costs are defined by equation 4.1.

Figure B.3. Asymmetric Trade Costs and Investment Adjustment



Note: The figure shows the results of estimation of equation (29). TIV estimates in square-black, OLS in circle-green. Controls: log terms of trade, log REER, log relative expenditures, four-year lags of all controls and dependent variable. Real GDP in log values. 90% confidence intervals; standard errors clustered by country. Inventories changes are calculated as total aggregate investment minus gross fixed capital formation. Confidence intervals are constructed at the 90% level, and standard errors are clustered at the country level.

Figure B.4. Trade Cost Changes Distribution



Note: The figure shows the yearly change on common trade costs (blue), and the assymetric trade costs (red).